

Nonstandard Educational Careers and the Returns to Vocational Education^{*}

Moritz Mendel[†]

December 3, 2025

This paper examines how flexibility in educational pathways, specifically the option to enter higher education after completing a vocational program, affects students' outcomes. I estimate a dynamic model of educational choices and outcomes using administrative data from the Netherlands. The model features a choice between general and vocational education at age 16 and allows for multiple subsequent educational transitions, including entry into university after completing a vocational program. The results show that the option to enter higher education after vocational training substantially increases graduation rates and average wages. The main mechanism behind this result is that students face considerable uncertainty about their future outcomes when making educational decisions at age 16. Allowing flexibility in educational pathways enables students to adjust their choices as they learn more about their abilities and preferences, leading to improved overall outcomes.

Keywords: vocational education, sequential choices, uncertainty

JEL code: J24, J31, I26, D80

^{*} I thank Teresa Backhaus, Manudeep Bhuller, Daniel Carvajal, Rik Dillingh, Thomas Dohmen, Hans-Martin von Gaudecker, Cormac O'Dea, Matti Sarvimäki, Paul Schäfer, Alessandro Toppeta, Paul Verstraten, Christian Zimpelmann, Maria Zumbuehl and various seminar audiences for helpful comments. I thank the CPB for hosting me for a month and for various helpful comments. I gratefully acknowledge funding by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) through CRC TR 224 (Project C01) and the Research Training Group "The Macroeconomics of Inequality". I also acknowledge funding by the Research Council of Finland (358924, 358946).

[†] Aalto University. moritz.mendel@aalto.fi

1. Introduction

The option to enroll in vocational training, rather than continuing with general education, offers an alternative pathway for individuals who are less academically inclined and unlikely to pursue university education. Several studies show that vocational training can improve labor market outcomes for parts of the population ([McNally et al., 2022](#)). However, expanding vocational education systems involves a fundamental trade-off. While vocational education may benefit some individuals, it may also divert others who would have had high returns from following the general track and enrolling in university. [Matthewes and Ventura \(2022\)](#), for example, provide evidence for this diversion effect in the UK.

This paper examines how flexibility in educational pathways, specifically the option to enter higher education after completing a vocational program, affects students' outcomes and the size of the diversion effect of vocational education. Most prior research treats the choice between vocational and general education as a static, binary decision and typically ignores subsequent educational choices. In contrast, I show that in the Netherlands, a substantial share of students who initially choose vocational education later enroll in tertiary programs. If students can enroll in university after completing vocational programs, then the extent to which vocational tracks divert students from university depends not only on how effectively the education system allocates students between general education and vocational programs. It also depends on how well vocational programs prepare students for higher education and on institutional barriers, such as entry requirements, that make transitions from vocational programs to university more difficult.

The opportunity to pursue university studies after completing vocational education, which I refer to as an alternative path to university, may mitigate diversion effects by allowing individuals to reverse or adjust earlier decisions. This flexibility is particularly important for students who may initially lack the grades, financial resources, or motivation to commit to university but later benefit from the availability of alternative pathways. At the same time, these alternative paths to university could exacerbate diversion effects if they encourage early choices that ultimately hinder university enrollment.

To study how the availability of alternative paths to higher education shapes individuals' outcomes, it is necessary to understand how students select into general education and vocational programs initially, how these choices shape their future options, and how students decide about university enrollment after completing a vocational program. Policies that modify either initial tracking or the availability of alternative pathways to university inevitably shift the composition of students entering

higher education through each route. As a result, it becomes challenging to isolate the effect of choosing one path over another. To address this challenge, I introduce a dynamic discrete choice model in the spirit of [Keane and Wolpin \(1997\)](#) that features both general and vocational education and allows for nonstandard educational careers. Estimating this model enables me to analyze how initial tracking and the availability of alternative pathways to higher education jointly shape students' outcomes.

The model follows graduates of vocational school in the Netherlands¹. Dutch students are assigned to different schooling tracks at age 12 based on test scores and teacher recommendations. Vocational schools enroll students with the lowest assessed academic potential and prepare them for vocational programs after graduation at age 16. The model begins when students graduate from vocational school and must choose between enrolling in applied high school (HAVO) or pursuing a vocational program (MBO). Individuals can enter applied university² either after completing applied high school, which is the direct path, or after completing a vocational program, which is the alternative path. Vocational programs take longer than applied high school and provide no explicit preparation for higher education. As a result, individuals who take the alternative path enter applied university at an older age and are potentially less academically prepared. Graduates of vocational school are close to the margin between pursuing higher education and entering the labor market after completing a vocational program, and they disproportionately come from lower-income families. Because they choose between rejoining the direct path to applied university through applied high school and pursuing a vocational program with the option of entering applied university later, they provide an ideal setting for examining the importance of alternative paths to university.

I leverage data on schooling careers, enrollment, and wage outcomes to estimate model parameters. In addition, I introduce a novel identification argument that relies on observed differences across schools to identify the incidence of unobserved heterogeneity in the model. The motivation for including this variation is that transition rates to applied high school in the first period differ substantially across vocational schools, even when conditioning on observed characteristics. While there is some anecdotal evidence that schools differ in their transition rules, it remains unclear how much of the observed differences across schools reflect selection versus genuine differences in rules. I show that variation across school types provides partial identification of the incidence of unobserved heterogeneity in the model, similar

¹ In particular, I focus on graduates of the theoretical branch of vocational school (VMBO-T). See [2.1](#) for more details.

² The Netherlands has two types of higher education institutions: academic and applied universities.

to using causal effects in the estimation process. The intuition behind the argument is that future conditional choices across schools provide information on whether observed differences are driven by exogenous variation in transition rules or by selection on unobservables.

I validate the model by comparing the model predictions to two reforms that changed rules related to the transition from vocational school to high school and from vocational programs to applied university, respectively. The model predictions line up well with the observed changes in choices following the reforms.

The estimated parameters show that lifetime earnings returns to applied university vary substantially across the population. For some individuals, the returns are negative because the higher earnings later in life do not compensate for the income lost while attending applied university. For most of the population, however, completing an applied university degree leads to a significantly higher lifetime income. At the same time, students face considerable dropout risk when entering applied university, in particular, individuals with low grades have only about a 50% chance of graduating.

I run two types of counterfactual simulations using the estimated model. First, I simulate two policies that expand access to the direct path to applied university by making it easier for vocational school graduates to enter applied high school at the beginning of the model. While these policies lead to a substantial increase in applied high school enrollment, they have only a modest effect on applied university graduation rates due to the high dropout risk among compliers. Overall, these policies would have a small positive effect on wages at age 40, as they increase the number of applied university graduates and raise average labor market experience among them, since more students take the shorter direct path to applied university. The results show that easing tracking rules at the beginning of the model by expanding access to applied high school helps some students but also draws in others who are not adequately prepared, while some who would succeed in applied university at a later stage still choose vocational programs.

In the second counterfactual exercise, I simulate two policies that restrict access to the alternative path to applied university. The first policy increases the duration of applied university for people who enter after completing a vocational program by two years, and the second policy completely eliminates the option to enter applied university after completing a vocational program entirely. Both policies are associated with substantial declines in applied university graduation rates and average wages at age forty. Removing the alternative path to university entirely would lead to a 15 percentage point decline in applied university graduation rates and a 6 percentage point reduction in average wages at age 40.

The main explanation for the large effect of alternative paths to higher education is that many people face substantial uncertainty when deciding between vocational programs and high school at 16. Allowing them to reconsider their initial choice later in life improves outcomes significantly. One of the key findings driving this result is that the penalty for entering applied university after completing vocational training instead of high school is relatively small.

Finally, I compare the returns to enrolling in a higher vocational program for people with different alternative options in the baseline scenario and the scenario without an alternative path to university. This exercise is motivated by recent studies, analyzing returns to vocational education and community colleges by the next-best alternative ([Matthewes and Ventura, 2022](#); [Mountjoy, 2022](#)), and documenting substantial negative returns to people being diverted from a more academic option. I document that the diversion effect becomes substantially larger in the counterfactual without an alternative path to university, highlighting the importance of future choices and the institutional environment in shaping long-run outcomes of graduates of vocational training.

The results of the structural model yield two crucial insights. Allowing individuals to pursue vocational training at age 16 instead of continuing high school improves outcomes for individuals who face considerable dropout risk and have only modest returns to applied university. At the same time, it diverts some individuals who would have high returns from higher education but do not yet know they want to study at 16. Providing flexibility between different education options allows one to reap the benefits of providing different options while keeping the losses due to wrong choices under uncertainty at a young age limited.

This paper contributes to several branches of the literature. The first is a growing literature studying how vocational programs shape individuals' outcomes. [McNally et al. \(2022\)](#) review the literature on the returns to vocational education and training, which includes studies using a wide range of methods in different institutional settings. [Silliman and Virtanen \(2022\)](#) find positive wage returns to entering the vocational track for middle school graduates in Finland who choose between general and vocational education. [Birkelund and van de Werfhorst \(2022\)](#) find that entering the vocational track increases employment probabilities in Denmark. In contrast, [Hanushek et al. \(2017\)](#) show that vocational education in Germany is associated with a higher unemployment risk later in life relative to general education. [Matthewes and Ventura \(2022\)](#) study returns to vocational education in the UK by next-best alternative and find substantial negative returns for individuals who would otherwise enroll in university. [Bertrand et al. \(2021\)](#) analyze a reform of vocational education in Nor-

way that made it easier for vocational graduates to access tertiary education. They find that the reform increased earnings among vocational graduates but had only limited effects on college enrollment. [Adda and Dustmann \(2023\)](#) estimate a model of wage formation for a population of German workers with vocational training and without any post-secondary education. They find that workers with vocational training accumulate cognitive-abstract skills faster, which has important consequences for their future job tasks and wages. [Eckardt \(2023\)](#) investigates the consequences of uncertainty in vocational program choice and quantifies the costs of occupational mismatch.

The mixed evidence in this literature highlights the importance of heterogeneity and institutional details in shaping returns to vocational education. In this paper, I estimate a fully structural model, which requires stronger assumptions but sheds light on the mechanisms that drive choices and outcomes among students entering vocational education. In addition, I explicitly account for students' future educational choices after they enroll in vocational programs. This enables me to examine how tracking rules and the availability of alternative pathways to university jointly shape later outcomes. Taken together, this approach allows me to document how the returns to vocational training vary across the population, how the expected returns to vocational versus university education depend on academic risk and ex-post wage outcomes, and how these returns are influenced by the availability of alternative routes to university.

Another closely related literature examines returns to community colleges and GED certificates in the US. [Mountjoy \(2022\)](#) analyzes returns to community colleges against different next-best alternatives and finds positive returns for people who would otherwise not attend college but negative returns for people who would otherwise attend a four-year college. One particularly close study is [Zhu \(2025\)](#), which estimates a dynamic model of education choices where individuals decide between community colleges and regular colleges and evaluates how free community college would promote social mobility. Another related study is [Heckman et al. \(2012\)](#) analyze how the availability of the GED certificate affects high school dropout decisions and labor market outcomes. They show that the availability of the GED certificate increases dropout rates and has detrimental effects on labor market outcomes and educational attainment. This study shows that in some cases, the availability of alternative paths to higher education can have negative diversion effects, outweighing the positive effects of flexibility. While the setting in the United States is different from the one studied in this paper, many of the general dynamics are similar across the two settings.

I also contribute to a literature using dynamic discrete choice models to study the effect of education policies when people make sequential education choices under uncertainty. [De Groote \(2025\)](#) and [Fu and Mehta \(2018\)](#) use dynamic discrete choice models to study the implications of educational tracking on students' outcomes. Their work focuses on more permeable tracking systems than the binary choice between vocational and general education studied in this paper, and does not account for alternative paths to university. [Stinebrickner and Stinebrickner \(2012\)](#), [Proctor \(2022\)](#), and [Arcidiacono et al. \(2016\)](#) emphasize the role of learning about one's own ability via noisy signals such as grades. They find that uncertainty about one's ability drives common phenomena such as dropout or re-enrollment. While the model in this paper does not feature learning explicitly, uncertainty plays an important role in shaping people's education choices in general and the importance of alternative paths to university in particular. A number of other studies have used additional data or variation from policy changes or RCTs to identify the incidence of unobserved heterogeneity in dynamic models of education choice. [Todd and Wolpin \(2023\)](#) provide a survey of structural papers that have used RCTs or quasi-experimental research designs to identify unobserved heterogeneity or to validate structural models. [Joensen and Mattana \(2021\)](#) use a reform to student loan policy in Sweden to estimate a dynamic model of education choice, loan uptake, and work choices. [Tincani et al. \(2023\)](#) use results from a randomized control trial to identify unobserved heterogeneity in a dynamic model of effort choice in high school. Since there is no exogenous variation that shifts all the relevant choices and outcomes in this paper, I take a different approach and show how to use observable variation across schools to identify unobserved heterogeneity.

The rest of the paper is organized as follows. Section 2 provides stylized facts and institutional details about the Dutch education system. Section 3 introduces a dynamic model of education choices. Section 4 introduces an identification argument and summarizes the estimation procedure. Section 5 discusses the main model results. Finally, I conclude in Section 6.

2. Setting and Stylized Facts

In this section, I explain relevant features of the Dutch education system, show stylized facts motivating the subsequent analysis, and summarize all the options that graduates of vocational school have.

2.1. Tracking in the Netherlands

The Dutch education system separates individuals at age twelve based on grades and teacher evaluations and sends them to different secondary schools. Each school sets a different focus and prepares for a different post-secondary education. The vocational schooling track (VMBO) receives individuals with the lowest assessed academic potential, takes three years, and prepares students for vocational programs. This paper will refer to the vocational schooling track as vocational school. Vocational programs (MBO) prepare individuals for particular occupations and take two to five years. The mid-level track (HAVO) prepares individuals for applied university and takes five years. I refer to this track as applied high school in this application. Higher education in the Netherlands differentiates between applied universities, which are more practical, and academic universities. The academic track (VWO) prepares individuals for academic university and takes six years. A bachelor's degree at an applied university takes four years and a bachelor's degree at an academic university takes three years. I will describe different career options for graduates of vocational school in section 2.5. I will abstract from academic university and master's programs in this context, as most of the graduates of vocational school never enroll in either.

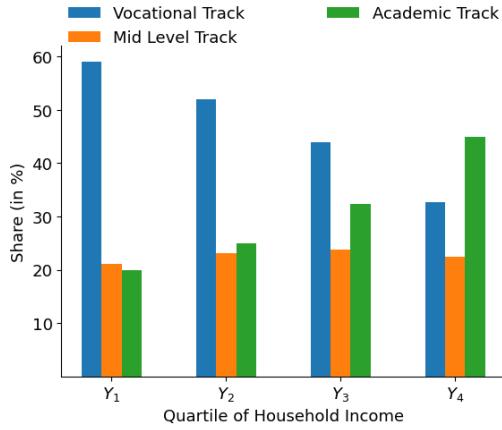
2.2. Data

I use Dutch administrative records to follow graduates of vocational schools. I combine information on educational careers, grades, the economic situation of their parents, school characteristics, place of residence, and future labor market outcomes. I will focus on graduates of vocational school and their future outcomes for the structural model.

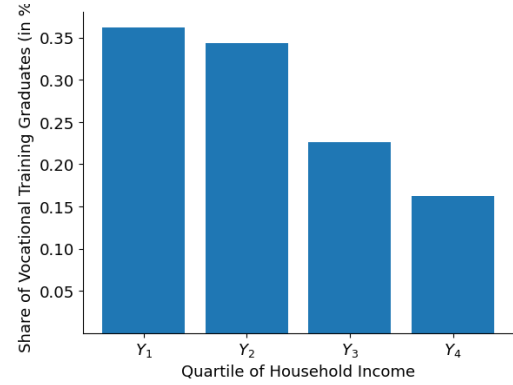
2.3. Sample Selection

I make several sample restrictions to make the analysis tractable.

1. I remove all people who enroll in the same education program after they have already graduated from another course in the program
2. I remove everyone who interrupts their schooling career for more than 2 years. This is to deal with people who enroll in university late.
3. I remove all people with inconsistent education careers



(a) Tracking across parental income quartiles



(b) Alternative path to university across parental income quartiles

Figure 1: Descriptive statistics on tracking and alternative paths to university

Note: This figure shows descriptive statistics about tracking and the prevalence of the alternative path to university. The first panel shows the proportion of individuals entering different secondary school tracks after primary school at age 12 by parental income quartile. The figure is based on the dataset described in 2.2 and contains data from 2008-2010. The second panel shows the proportion of university graduates that have completed vocational training before entering higher education by parental income quartile. University graduates include everyone with at least an applied university bachelor's degree. Individuals with an academic university bachelor's degree or any master's degree are also included. The figure is based on the dataset described in 2.2 and contains data from 2008-2010.

2.4. Descriptive Statistics

Initial tracking: Figure 1 summarizes track choice after primary school by parental income. The figure shows that the majority of people visit a vocational school after primary school. Students from low-income backgrounds are most likely to be selected for vocational school. Track assignment is decided by teacher evaluations and a centralized test individuals take at the end of primary school. Grade differences at the end of primary school can explain a substantial part of the differences in track choice. Zumbuehl et al. (2022) show that individuals from low-income backgrounds, however, receive lower track recommendations even after controlling for grades and cognitive skills. The misallocation is thus potentially worse among individuals from low-income backgrounds than among their higher-income peers.

Alternative paths to applied university: I now consider all individuals who at least hold an applied university bachelor's degree. The second panel of Figure 1 shows the proportion of university graduates that have completed vocational training before. The figure shows that almost a quarter of all university graduates have completed vocational training before entering higher education. Graduates of vocational education are older and have received less academic education when they consider entering university. This shows that the alternative path to university is a well established educational pathway in the Netherlands. Conditional on reaching a tertiary degree, individuals from low socioeconomic backgrounds are twice as likely to have entered higher education after vocational training. Entering university after finishing voca-

tional training is thus a well-established career in the Netherlands that is particularly important for individuals from low-income backgrounds.

Observed wage differences: Average wages differ substantially across educational attainment. Figure A.4 in the appendix shows the average yearly income for people with different relevant degrees. Applied university graduates show substantially larger wage growth between ages 30 and 40 than people with other degrees. People who hold an applied high school degree start out lower than graduates of vocational programs, but catch up at age 40. It is important to point out that the vocational category includes individuals with different levels of vocational training. Graduates of a higher vocational program (MBO4) are likely to earn more than other people who fall under that category.

2.5. Pathways to university

In this section, I present all possible future pathways for graduates of vocational school. From this point onward, I focus on graduates of the theoretical branch of vocational school. Vocational education is divided into four branches, and the theoretical branch admits students with the highest assessed academic ability among them. Students in this branch are closest to the margin between entering applied university and entering the labor market after completing a vocational program. Students in the lower branches are unlikely to enroll in applied university, whereas students who are initially tracked into higher academic tracks are less likely to ever attend a vocational program. The theoretical branch of vocational school is therefore the most relevant population for understanding the importance of alternative paths to university. In 2010, roughly one quarter of all secondary school graduates came from the theoretical branch of vocational education³. Consistent with the stylized facts mentioned earlier, individuals from low-income backgrounds are overrepresented in this group.

Figure 2 illustrates the possible pathways that vocational graduates can pursue after graduation. After finishing vocational school, individuals may enroll in different vocational programs or switch to the schooling track that prepares them for applied university, which I refer to as applied high school. Once individuals graduate from applied high school or a higher vocational program, they can enter applied university. If they hold a lower vocational degree, they can pursue a higher vocational degree to enter applied university in the third period. Naturally, Figure 2 includes some simplifications. In particular, I leave out possibilities that are negligible empirically. While lower vocational programs contain options beyond MBO3, most graduates of the the-

³ See [CBS opendata](#)

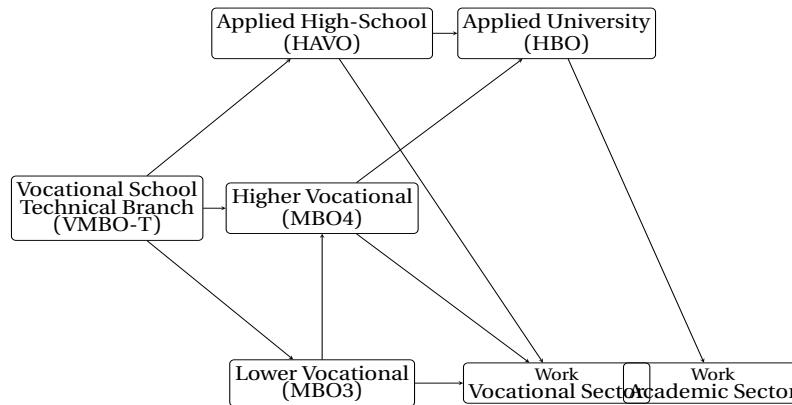


Figure 2: Pathways for graduates of vocational school

Note: This figure summarizes educational careers individuals can pursue after graduating from a vocational school.

oretical branch choose the latter. Individuals could switch to an academic high school (VWO) after finishing applied high school (HAVO), and they could change to an academic university during their studies at an applied university. I abstract from both of these options as they are chosen infrequently. Finally, individuals can enroll in a master's degree after finishing applied university. I also abstract from this choice and treat individuals with applied university master's and bachelor's degrees equally.

Transition to applied high school: The transition to applied high school is not organized centrally. Applied high schools have employed their own rules for admitting students from vocational school ([Van Esch and J., 2010](#)). The number of individuals that transfer to high school from a particular vocational school thus varies by the specific rules that high schools in the area use and by the amount of assistance that the school offers students for their transition to high school.

3. A Model of Further Education

I now introduce a structural model of education. I will first introduce all the components of the model, then show how to solve the model.

3.1. Sample and decision tree

The model is based on the summary of pathways introduced in Figure 2. Individuals can first choose between a lower vocational program, a higher vocational program or applied high school. After that, they can enter applied university after applied high school or after graduating from a higher vocational program. A higher vocational program takes longer than applied high school and contains less preparation for applied

university. At each stage of the model students can also decide to leave education and start working. The sample of individuals the model is estimated with consists of all graduates of the theoretical branch of vocational school, as described in the last section. In addition to the selection criteria described in Section 2, I restrict the sample to people who graduate from vocational school in the years 2009 and 2010. I introduce this additional restriction to ensure that the rules governing transitions to applied high school do not change over time within a given vocational school. Moreover, I abstract from part-time work and only use full-time work spells to estimate wage processes.

3.2. Model organization and decision period

Contrary to prior dynamic discrete choice models of education, individuals do not make a new decision each year. I choose this alternative way of specifying the model to reduce the computational complexity. After individuals enroll in a particular education program, they stick with this decision for a potentially stochastic number of years until they either graduate or fail to do so. A spell denotes the years an individual spends in a particular education due to their prior decision. Once the current spell is over, they make a new decision based on their current state. I thus distinguish between periods and decision periods in the model. A period $t \in \{0, 1, 2, 3, \dots, 13\}$ denotes the number of years that have passed since the onset of the model. A decision period $\tau \in \{0, 1, \dots\}$ represents the number of choices that the individual has already taken. Using decision periods allows me to substantially reduce the number of states because I do not have to include experiences for each choice in the state space.

3.3. States and fixed heterogeneity

Each individual is characterized by fixed characteristics and dynamic states. Fixed characteristics include academic potential G , latent type θ , parental income Y , and school type Z . Academic potential G is the probability of entering applied high school based on a large set of observed characteristics. The most important characteristics are grades, but the measure also includes subject choice or the number of exams chosen. The score is then obtained by using a gradient boosting regressor to predict the probability of entering applied high school based on these characteristics. Y denotes the quartile of parental household income. School type Z denotes the type of transition policy in the individual's school. This variable captures that transitioning to applied high school after graduating from vocational school is easier from some vo-

cational schools than others. I obtain school type Z for school j as follows:

$$Z_j = q\left(\frac{1}{n_j} \sum_{i \in I_j} G_i - \hat{P}_j\right)$$

Where $\hat{P}_j$ is the observed transition rate to applied high school in school j , n_j denotes the number of students at school j , $\frac{1}{n_j} \sum_{i \in I_j} G_i$ is the predicted transition rate conditional on observables for school j , and q is a function that maps the school-specific deviation to its quartile in the distribution. The school type thus captures the variation in transition rates across schools that cannot be attributed to differences in observables across schools. Latent type θ is a latent factor that captures dependence between choices and outcomes not accounted for by observed characteristics. All fixed characteristics are assigned at the beginning of the model. The joint distribution of Y , Z , and G is assigned exogenously as observed in the data. The distribution of θ depends on all the other fixed states and is estimated with all other parameters. Dynamic states include age A , current level of schooling E , and lagged choice $d_{\tau-1}$. One state is a tuple that consists of all fixed characteristics and dynamic states as described in Equation 1. Individuals start the model at age 16.

$$s_\tau = (A_\tau, E_\tau, d_{\tau-1}, G, \theta, Y, Z) \quad (1)$$

3.4. Choices and timing

Let d_τ denote an individual's choice at decision period τ . At each decision period, an individual makes a choice. Afterward, the individual stays with that choice for a potentially stochastic number of periods. After the spell is over, the individual takes the next decision.

$C(s_\tau)$ maps a state into a set of admissible choices. This function is consistent with the decision tree above. An individual who has, for example, just finished a higher vocational program can either enroll in university or leave education and work. Moreover, individuals are not allowed to enroll in the same program repeatedly. This is why the lagged choice is part of the state space. Individuals decide between applied high school, higher vocational training, lower vocational training and work in the first stage. After that, the set of choices depends on their state. People are not allowed to enroll in education programs beyond a certain age such that everyone starts working by age 32 at the latest.

If individuals enroll in a particular schooling program, they are not guaranteed to finish it. Schooling programs are associated with varying levels of dropout risk and un-

certain length. Depending on their choice and the realization of academic risk, they will transition to a new stage. The stochastic function $T(s_\tau, d_{i,\tau})$ maps a state and a choice into a state at the end of the current spell.

Taking a decision thus has the following consequences. First, the transition function realizes and determines the state that an individual will end up in. Function $N(s_\tau, s_{\tau+1})$ determines all the states in between the state of departure and the state of arrival and $n(s_\tau, s_{\tau+1})$ is the number of states between s_τ and $s_{\tau+1}$. After that, the individual receives utility for each state and makes a new decision in the arrival state. Suppose the transition function, for example, determines that an individual enrolled in a higher vocational program will graduate within four years. In that case, the individual will receive utility for these four years and make a new decision after she graduates from the vocational program.

If an individual leaves education and starts working, the choice is terminal. Individuals receive the discounted lifetime income associated with their characteristics and final education.

3.5. Transitions and uncertainty

Individuals face two types of uncertainty in education: they can potentially drop out and not graduate from a particular education program, or they can graduate but with a delay.

Equation 2 shows the specification of dropout risk. $P(E_{\tau+1} = d_\tau)$ is the probability that an individual successfully graduates from the education program she enrolled in. The equations' coefficients are model objects estimated jointly with all other parameters.

$$\text{logit}(P(E_{\tau+1} = d_\tau | G, \theta, Y)) = \beta_{0,d}^R + \xi_{1,d}^R G + \xi_{2,d}^R \theta + \xi_{3,d}^R Y \quad (2)$$

Let $\min_d$ be the minimum years required to finish a degree. If an individual i completes a degree successfully, she faces a shifted Poisson process that determines the duration of her degree:

$$T_d^{E_{\tau+1}=d_\tau}(G, \theta, Y) \sim \text{Poisson}(\min_d, \beta_{0,d}^D + \xi_{1,d}^D G + \xi_{2,d}^D \theta + \xi_{3,d}^D Y). \quad (3)$$

If the individual drops out, she will still spend a stochastic number of periods in the education program. The length is determined by:

$$T_d^{E_{\tau+1} \neq d_\tau} \sim \text{Poisson}(1, \beta_0). \quad (4)$$

The exact parametrization differs between the programs and can be found in the appendix.

Agents additionally face taste shocks $\nu_{i,\tau}(d)$ to their utility. Taste shocks are modeled as an extreme value type one distribution. They are independent and identically distributed across all individuals and all choices.

3.6. Wages and nonpecuniary preferences

Wages are modeled as two separate equations for individuals with higher education diplomas and individuals without. Once students enter the labor market, they receive income for the rest of their lives. I assume that everyone works full-time after they leave school. Let k_t be work experience at time t and let E^C be an individual's combination of degrees. Log wages for the vocational sector are specified in Equation 5. Log wages in the vocational sector depend on experience, age, parental income, academic potential, type, highest degree completed, and highest degree completed interacted with experience.

$$w_v(E, A_t, k_t, G, \theta, Y) = \beta_{0,v}^W + \beta_{1,v}^W E + \beta_{2,v}^W k_t + \beta_{3,v}^W k_t^2 + \beta_{4,v}^W A_t + \beta_{5,v}^W k_t E \\ + \beta_{6,v}^W k_t^2 E + \xi_{1,v}^W G + \xi_{2,v}^W \theta + \xi_{3,v}^W Y + \xi_{4,v}^W k_t \theta + \epsilon_{v,t} \quad (5)$$

Log wages in the academic sector are modeled separately in equation 6. I use a different specification for academic wages to allow for a flexible form of the applied university wage premium. They depend on experience, age, parental income, ability, type, and educational career.

$$w_a(E^C, A_t, k_t, G, \theta, Y) = \beta_{0,a}^W + \beta_{1,a}^W E^C + \beta_{2,a}^W k_t + \beta_{3,a}^W k_t^2 + \beta_{4,a}^W A_t + \xi_{1,a}^W G + \xi_{2,a}^W \theta \\ + \xi_{3,a}^W Y + \epsilon_{a,t} \quad (6)$$

Similar to [Keane and Wolpin \(1997\)](#), every choice is associated with nonpecuniary utility that is measured on the same scale as wages. I allow nonpecuniary returns $F(s, d_t)$ to depend on parental income, type, and dynamic characteristics such as experience or age. Academic potential is only part of nonpecuniary rewards for high school, where higher grades may be associated with lower transition costs. Additionally, I include transition costs to applied high school $T(Z)$, to capture differences in transition costs across school types. I assume that the differences in rules across schools can be captured by a school-specific constant shift in the non-pecuniary utility associated with choosing applied high school in the first period. Equation 7 shows the flow utility associated with being in an education program d in state s . Since rules make only a difference for applied high school enrollment in the first period, the last

term in this equation only enters for applied high school.

$$U_d(s) = F_d(s) + T_d(Z) \quad (7)$$

Equation 8 shows the utility associated with work in state s . Depending on their final education, individuals receive wages from either the vocational or academic wage equation.

$$U_w(s) = F_w(s) + e^{w(s)} \quad (8)$$

The coefficients of wage equations, nonpecuniary returns to choices, and transition costs are all model objects that are estimated. Equation 9 denotes the discounted lifetime utility from working if an agent starts working in state s at time t_s . Let $s_{w,t}(s)$ be a deterministic function that maps a state s at time t_s into the corresponding states at time t . This function updates time and experience to calculate the discounted lifetime utility from working. The term β is the discount factor and estimated along with the other parameters.

$$TU(s) = \sum_{t=t_s}^T \beta^{t-t_s} U_w(s_{w,t}(s)) \quad (9)$$

3.7. The agent's problem and solution algorithm

Expected utility is the weighted average over all possible paths a decision could lead to. One needs to sum over all states that could be reached from a particular state choice combination. Let $R(s_\tau, d_{i\tau})$ be the range of potential outcomes one can reach from state s_τ and decision $d_{i\tau}$ and let $P_{s_\tau, d_{i\tau}}(s_{\tau+1})$ be a probability distribution over the range of outcomes. Equation 10 shows the optimization problem of an individual in the model at state s_τ . Note that if the decision is to work then $n(s_\tau, s_{\tau+1})$ is zero and the whole term is equal to $TU(s)$ as specified in Equation 9.

$$\max_{d \in C(s_\tau)} \left\{ \sum_{s_{\tau+1} \in R(s_\tau, d)} P_{s_\tau, d}(s_{\tau+1}) \left[\sum_{s \in N(s_\tau, s_{\tau+1})} \beta^{n(s_\tau, s)} U_d(s) + \beta^{n(s_\tau, s_{\tau+1})} V(s_{\tau+1}) \right] + \nu_{i, \tau}(d) \right\} \quad (10)$$

I solve the model by backward induction. Let $V(s)$ be the expected continuation value from reaching state s , let $V(s, d)$ be the expected continuation value from choosing d in state s , and let $V(s, d, \hat{s})$ be the expected continuation value of choosing d in state s and reaching $\hat{s}$. Also I refer to a state as terminal if the only choice available is work. At the last time period where people can take decisions, all states are terminal. To solve this model, I proceed as follows. I start with the highest age at which agents can make decisions in the model and repeat the following steps for all ages down to age 16.

1. Collect all possible state choice combinations (s, d) of age t
2. For all combinations of states and work choices, assign the terminal utility

$$V(s, w) = TU(s)$$

3. For all non-terminal combinations:

- a) Collect all reachable states $\hat{s} \in R(s, d)$ and their probability $P_{s,d}(\hat{s})$
- b) Collect the expected continuation value from reaching $\hat{s}$: $V(\hat{s})$
- c) Now combine the expected continuation value with the flow utility on the path from s to $\hat{s}$:

$$V(s, d, \hat{s}) = \sum_{\tilde{s} \in N(s, \hat{s})} \beta^{n(s, \tilde{s})} U_d(\tilde{s}) + \beta^{n(s, \hat{s})} V(\hat{s})$$

- d) Get the continuation value of (s, d) by taking the expected value over $\hat{s}$:

$$V(s, d) = \sum_{\hat{s} \in R(s, d)} P_{s,d}(\hat{s}) V(s, d, \hat{s})$$

4. Now get $V(s)$ by getting the expected value of the maximum of $V(s, d)$: $V(s) = E[\max\{V(s, d)\}] = \sigma \log(\sum_d e^{\frac{V(s, d)}{\sigma}})$ where σ is the scale of the extreme value taste shocks.

4. Estimation and Identification

This section will summarize the estimation procedure and the variation I use to identify the model parameters. In the first part of this section, I will present an argument of how variation in conditional choices and outcomes by school type helps to identify the distribution and impact of unobservable characteristics in the model. In the second part, I will present the whole set of moments and summarize how they are informative about the parameters of the model.

4.1. Identification and Variation Across Schools

In this section, I will show how the variety of conditional choices and outcomes implied by the model earlier allows me to extract further information from variation across school types. I will first describe variation across school types and build some

intuition about how this variation is informative about unobserved heterogeneity. Then I will show how variation across school types provides partial identification for the distribution of unobserved heterogeneity without invoking any functional form assumption except for the discrete nature of unobserved heterogeneity. Finally, I will discuss how this approach can be combined with further assumptions invoked by the model introduced earlier to pin down the distribution of unobserved heterogeneity.

Variation across schools As described in Section 2.5, the transition from vocational school to applied high school differs substantially across schools. Figure A.1 shows that the lowest school type sends around 8 percent of its graduates to applied high school, while the highest school type sends around 35 percent. While there is some evidence that schools use different rules when deciding whom to admit to applied high school, it is unclear to what extent differences in transition rates are driven by differences in rules $T(Z)$ or selection across schools $corr(Z, \theta)$. Conditional choices and outcomes across school types, however, provide information that helps to understand selection patterns across school types. This information can then be used to disentangle differences in rules from differences in selection and to identify the incidence of unobserved heterogeneity in the model.

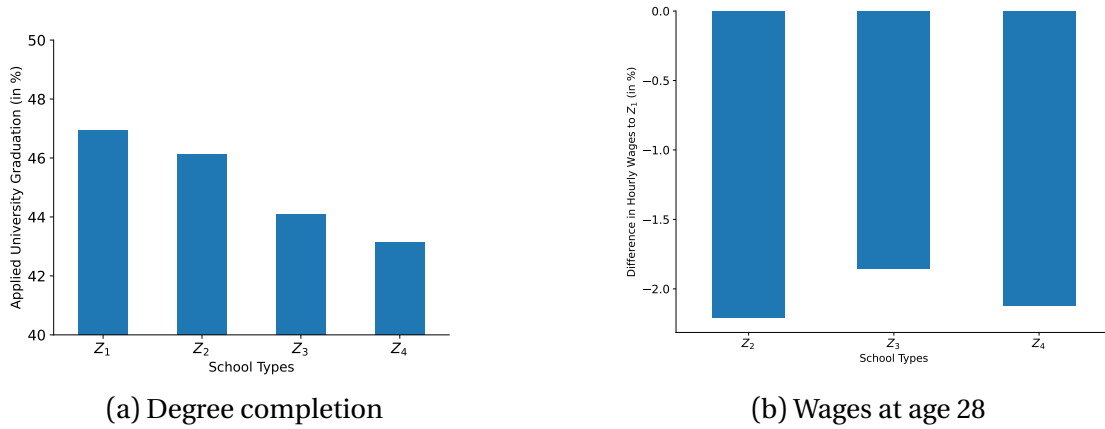


Figure 3: Variation across school types

Note: This figure shows differences in conditional choices and outcomes across school types. Schools are grouped into four types based on their covariate-adjusted transition rate to applied university as outlined in Subsection 3.3. The first panel shows differences in the probability of graduating from applied university, conditional on not having chosen the direct path to university for the highest quartile of academic potential. The y-axis represents the proportion of graduates. The different school types are represented by different bars on the x-axis. The second panel shows differences in wages at age 28, conditional on having graduated from applied university via the alternative path. Wages are expressed relative to the school with the highest transition rate to applied high school. The different school types are represented by different bars on the x-axis.

Figure 3 shows conditional outcomes across school types to build intuition for how this variation is informative about selection across schools. The first panel shows the probability of graduating from applied university conditional on not having chosen the direct path to university for the highest quartile of academic potential. The figures show that for the highest academic potential group, schools with lower initial transition rates to applied high school have a higher probability of graduating

from the alternative path. The second panel shows wages conditional on having graduated from applied university via the alternative path. The plotted differences are fixed-effect coefficients of a regression of wages on school type, parental income, and academic potential. The figure shows that wages are higher for people from schools with lower transition rates to applied high school. Both of these patterns suggest that conditional on not choosing the applied high school in the first period, people from schools with lower transition rates are positively selected on unobserved characteristics that affect wages and graduation rates. If differences in rules across schools drive differences in transition rates and there is otherwise no selection into vocational schools, such a pattern should arise. If on the other hand selection into schools were the only driver of differences in transition rates, we would expect the opposite pattern.

Partial identification from cross-school variation In this section, I argue that adding variation in conditional choices and outcomes across school types provides partial identification for the distribution and incidence of unobservables similar to added causal effects derived from a natural experiment or RCT outside of the model. Given a candidate distribution of latent types, I derive an overidentified system of equations that relates the distribution of latent types across school types and choice margins to the observed moments across school types and choice trajectories and the candidate distribution of latent types. The argument is inspired by the analysis in [Kasahara and Shimotsu \(2009\)](#). The choice trajectory refers to whether individuals choose the direct path to university, the alternative path to university, or neither path. In the model introduced earlier, choosing the direct path means enrolling in applied high school in the first period, choosing the alternative path means enrolling in applied university after completing a higher vocational program, and choosing neither path corresponds to entering the labor market after vocational school or after graduating from a vocational program. I begin by introducing some additional notation:

- $c \in \{00, 01, 10\}$ is the choice trajectory of an individual. 10 indicates the direct path to university, 01 indicates the alternative path to university, and 00 indicates no university at all.
- y, g, g^{hs}, a, p^{lv} are outcomes (earnings, dropout at applied university, dropout at applied high school, age at graduation, proportion choosing the lower vocational program)
- P^{lv}, P^{01}, P^{10} refer to choice probabilities (lower vocational program, alternative path to university, direct path to university)

Each outcome, choice, and proportion can be indexed by school type z , treatment trajectory c , and latent type θ . Every combination of these indices defines the conditioning set for the corresponding variable. The outcome $y_{z,c,\theta} = E[y \mid z, c, \theta]$ is the conditional expectation of y given school type z , choice trajectory c , and latent type θ . The outcomes are defined to match the moments included in the estimation. A candidate distribution of latent types can be written as

$$\mathcal{F}_\theta = (y_{c,\theta}, a_\theta, g_\theta, g_\theta^{hs}, p_\theta^{lv}, p_\theta^d, p_\theta^a, P(\theta)).$$

In total, the candidate distribution of latent types contains ten parameters for each latent type. Note that in the model introduced earlier, the actual outcome of an individual depends on the final schooling degree rather than the choice trajectory. I index outcomes by choice trajectory rather than by final degree because the choice trajectories represent the central margins on which the school types operate. An outcome $y_{\theta,c}$ therefore represents the average outcome for type θ , taking into account dropout risk and the implied distribution of final degrees. In addition, let $P(z, c, \theta)$ denote the mass of individuals with school type z , treatment trajectory c , and latent type θ , expressed as a proportion of the full population. In the data, we can observe outcomes conditional on school type and treatment trajectory, such as $y_{z,c}$ and a population distribution over school types and choice trajectories $P(z, c)$.

I now derive a set of equations that relates $P(z, c, \theta)$ to both observed objects and the candidate distribution of latent types. I abstract from observable characteristics for ease of notation, but the notation can be extended in a straightforward way to include conditioning on observables. The full system of restrictions is shown in Equation 11. The first two equations in the block 11 enforce mass balance across latent types, choice trajectories, and school types. Equation 11a states that for each combination of z and c , the mass across latent types $\sum_\theta P(z, c, \theta)$ must be equal to the observed mass of individuals with school type z and choice trajectory c . Equation 11b states that for each latent type θ , the mass across all school types and choice trajectories $\sum_c \sum_z P(z, c, \theta)$ must correspond to the population share of type θ . Equations 11c–11g relate observed outcomes for each combination of choice trajectory and school type to weighted type-specific outcomes. For example, Equation 11c states that for each pair (z, c) the weighted outcome across types $\sum_\theta P(z, c, \theta) y_{c,\theta}$ must be equal to the observed wages $y_{z,c}$ multiplied by the observed mass of individuals with school type z and choice trajectory c . Finally, Equations 11g–11h relate observed choice probabilities to weighted type-specific choice probabilities. Equation 11g states that for each school type and latent type the fraction of individuals who choose the alternative path to university in the second period $\frac{P(z,01,\theta)}{P(z,01,\theta)+P(z,00,\theta)}$

must be consistent with the type specific choice probability P_θ^{01} . Equation 11h states that in each school type z , the overall fraction of individuals who choose the lower vocational program P_z^{lv} must be equal to the weighted type-specific choice probabilities P_θ^{lv} among individuals who do not choose the direct path to university.

$$\sum_{\theta} P(z, c, \theta) = P(z, c) \quad \forall c, z \quad (11a)$$

$$\sum_c \sum_z P(z, c, \theta) = P(\theta) \quad \forall \theta \quad (11b)$$

$$\sum_{\theta} P(z, c, \theta) y_{c,\theta} = P(z, c) y_{z,c} \quad \forall c, z \quad (11c)$$

$$\sum_{\theta} P(z, c, \theta) g_{c,\theta} = P(z, c) g_{z,c} \quad \forall c \in \{10, 01\}, z \quad (11d)$$

$$\sum_{c \in \{10, 01, 00\}} \sum_{\theta} P(z, c, \theta) a_{c,\theta} = P(z) a_z \quad \forall z \quad (11e)$$

$$\sum_{\theta} P(z, 10, \theta) g_{\theta,10}^{hs} = P(z, 10) g_{z,10}^{hs} \quad \forall z \quad (11f)$$

$$\frac{P(z, 01, \theta)}{P(z, 01, \theta) + P(z, 00, \theta)} = P_\theta^{01} \quad \forall \theta, z \quad (11g)$$

$$\sum_{c \in \{01, 00\}} \sum_{\theta} P(z, c, \theta) P_\theta^{lv} = P(z) P_z^{lv} \quad \forall z \quad (11h)$$

$$(11i)$$

Altogether this yields $11n_z + (n_z + 1)n_\theta$ linear restrictions and $3n_\theta n_z$ unknowns. In the case considered in this study with $n_\theta = 5$ latent types and $n_z = 4$ school types, we have 60 unknowns and 69 equations. For each candidate distribution of latent types, there exists a set of school-specific distributions of unobserved characteristics $P^*(z, c, \theta)$ that minimizes the weighted distance of this overidentified system of equations. Introducing variation across school types into the set of moments, therefore, incentivizes the estimation procedure to select a distribution of unobserved heterogeneity that is associated with a minimal school-specific distribution $P^*(z, c, \theta)$ that best aligns with the overidentified system in Equation 11. Equation 11 does not point identify the distribution of unobserved heterogeneity. If we were to treat the candidate distribution of latent types as an additional set of unknowns, the system would contain $60 + 50$ unknowns and only 69 equations. However, it provides partial identifying restrictions that help to narrow down the distribution of unobserved

heterogeneity in the model.

Combination with the model So far, I have not imposed any assumptions on the differences in rules across schools or on the functional form of the model. In this section, I show that placing structure on rule differences across schools and assuming additive separability between latent types and observable characteristics introduces additional restrictions that help to pin down the distribution of unobserved characteristics in the model.

Equation 11 allows for arbitrary patterns of transition rates across latent types and school types. In practice, however, it may be reasonable to assume that schools differ only along a one-dimensional margin, for example, because they apply different cutoffs for admitting students to applied high school. In the model introduced earlier, I parametrized differences in rules as a shift in the cost of choosing the direct path to university, denoted by $T(z)$. Let $m(T(z), P_\theta^{10})$ be a function that maps type-specific choice probabilities and the school-specific cost shift into type-specific choice probabilities in the first period. Such a function can be derived, for example, from the way $T(z)$ enters the nonpecuniary rewards in the model described earlier. Under these assumptions, I can add the additional restriction shown in Equation 12. This equation imposes that the type-specific probability of choosing the direct path at school z must correspond to the choice probability implied by the cost shift and the type-specific probability of choosing the direct path:

$$m(T(z), P_\theta^{10}) = P(z, 10, \theta) \quad \forall z, \theta \quad (12)$$

This condition yields another $n_z n_\theta$ nonlinear restrictions, which further help to pin down the distribution of unobserved heterogeneity in the model.

If observable characteristics and latent types enter outcomes in an additively separable way, one can derive a large set of additional restrictions that the distribution of unobserved heterogeneity must satisfy. In that case, the distribution of latent types would need to satisfy the constraints laid out in Equation 11 for each combination of observable characteristics, while the number of unknown parameters would increase only by the sum of all possible values of observable characteristics and their distribution across latent types. In the model introduced earlier, I assume that observable characteristics and latent types enter outcomes in an additively separable way. I also assume that observable characteristics are additively separable from each other, and I impose a functional form on the type distribution conditional on observable characteristics. Finally, I normalize the parameters associated with one latent type to zero in order to fix the location of unobserved heterogeneity. These parametric assumptions

introduce a large set of additional restrictions that help to pin down the distribution of unobserved heterogeneity in the model.

It is also reasonable to assume a systematic relationship between P_{θ}^{10} and P_{θ}^{01} and between $y_{z,10,\theta}$ and $y_{z,01,\theta}$, since each pair refers to different ways of choosing the same treatment. Invoking such assumptions reduces the number of parameters in the candidate distribution of latent types and sharpens the identifying restrictions. In the model introduced earlier, this is achieved by specifying one process that generates outcomes and nonpecuniary utility for applied university.

4.2. Estimation

I use indirect inference to estimate 117 parameters $\hat{\theta}$. Equation 13 shows the criterion function. I select the parametrization that minimizes the weighted squared distance between the specified set of moments computed on the observed M_D and the simulated data $M_S(\theta)$. I weight the statistics with a diagonal matrix W that contains the variances of the observed moments (Altonji and Segal, 1996). I use a package for the estimation of scientific models by Gabler (2022) for the optimization of the criterion function⁴.

$$\hat{\theta} = \arg \min_{\theta \in \Theta} (M_D - M_S(\theta))W^{-1}(M_D - M_S(\theta))' \quad (13)$$

Description of Moments: Table 1 provides an overview of all 344 statistics used in the model estimation. The enrollment percentage for a particular program indicates how many people have been enrolled in that program. Enrollment percentages are included for each quartile of parental income and each combination of school type and quartile academic potential. The final degree combination indicates all degrees an individual receives before starting work. If a person first graduates from a vocational program and then graduates from an applied university, her degree will be higher vocational & applied bachelor's. Final degree combinations are included for each quartile of parental income and each quartile of academic potential in vocational school. Furthermore, I include the last schooling age for all income quartiles and each combination of school type and quartile academic potential. The last schooling age is when an individual is done with education and starts to work. Since I do not allow re-enrollment, there is always one age at which individuals leave education. In practice, I allow individuals to take a gap of up to two years between spells, which will be part of the degree duration. Wage quartiles over time are wage quartiles for individuals by final degree at ages 30, 35, and 40. Conditional choices across school types

⁴ I use a global version of the BOBYQA algorithm within the package (Powell et al., 2009).

include the proportion of individuals enrolling and graduating from applied university across school types, conditional on their choices in the first period.

Finally, I match the coefficients of six separate wage equations. Let T^u denote the years someone needs to finish applied university. Let γ be year fixed effects. Equation 14 is estimated on a panel that includes all full-time individuals who left school without a bachelor's degree from the third period onward. Equation 15 is estimated on a panel that includes all full-time individuals who left school with a bachelor's degree from the sixth period onward. Both equations capture how wages depend on observable states featured in the model. Both include work experience k_t , academic potential G , and parental income Y . Equation 14 additionally includes the non-university degree of an individual and an interaction between the non-academic degree and work experience. This is either a higher vocational degree (MBO4), a lower vocational degree (MBO3), an applied high school degree (HAVO), or no degree after vocational school (VMBO-T). Equation 15 additionally includes a fixed effect for all non-university degrees that individuals have completed before entering university. Furthermore, it includes the years an individual took to finish her bachelor's degree. Both of these equations suffer from selection bias. Since the model explicitly models the selection process, they are still helpful for identifying wage components.

$$W_{v,t} = \alpha_{v,0} + \alpha_{v,1}E + \alpha_{v,2}k_t + \alpha_{v,3}k_t^2 + \alpha_{v,4}k_tE + \delta_{v,0}G + \delta_{v,1}Y + \gamma + \omega_{v,t} \quad (14)$$

$$W_{a,t} = \alpha_{a,0} + \alpha_{a,1}E^C + \alpha_{a,2}T^u + \alpha_{a,3}k_t + \alpha_{a,4}k_t^2 + \delta_{a,0}G + \delta_{a,1}Y + \gamma + \omega_{a,t} \quad (15)$$

Finally there are four equations that capture how wages depend on school type across different choice trajectories. Equation 16 is estimated on a cross-section of all full-time individuals in period thirteen. The equations contain school type fixed effects and fixed effects for quartiles of academic potential.

$$W_h = \alpha_{h,0} + \alpha_{h,1}Z + \alpha_{h,2}G + \alpha_{h,3}Y + \omega_h \quad (16)$$

The set of statistics is chosen to identify all components of the model. While the moments are used jointly, I will provide some heuristic arguments of how each category of moments aids identification. Coefficients of wage equations and wage quartiles pin down components of the wage equation. The discrepancy between enrollment and graduation in each program identifies academic risk. The distribution of final schooling ages pins down the distribution of degree duration. Non-pecuniary returns to work and education programs are pinned down by residual variation in choices across characteristics that wage returns cannot explain. The distribution of taste shocks is pinned down by variation in choices, holding all characteristics fixed.

Table 1: Summary of Moments Used in the Estimation.

Type of Moment	Number
I. Program enrollment by parental income	16
II. Degree combination by parental income, academic potential	56
III. Last schooling age by parental income	12
IV. Life-cycle wage quartiles by education	45
V. Coefficients of wage equations	29
VI. Program enrollment by school type x academic potential	64
VII. Conditional choices across school types	46
VIII. Conditional wages across school types	28
IX. Last schooling age by school type x academic potential	48

Note: This table summarizes all 344 moments used to estimate the model. The left column indicates a particular category of statistics, and the right column indicates the number of moments the respective category has. See Section 3.3 for an explanation of how academic potential is constructed.

Transition costs to high school by school type are identified by differences in choices and outcomes of individuals who chose not to enroll in high school. Latent types are identified in two ways. First, variation in conditional choices and outcomes across school types helps to identify the distribution of unobserved characteristics as argued in Section 4.1 above. Second, they are identified by all moments jointly as they introduce persistence in choices over time, which minimizes residual heterogeneity.

4.3. Model fit

Figure 4 briefly summarizes the model fit. A more detailed summary can be found in Section S3 in the [supplemental appendix](#). The first two panels show the fit of enrollment proportions and degree combinations for individuals with high academic potential. Both sets of simulated moments are closely aligned with their observed counterparts. The third and fourth panels show wage quartiles for individuals with an applied university degree at ages thirty and forty. The simulated moments closely match the observed moments.

4.4. Validation

This section presents two different validation exercises to assess the credibility of the assumptions imposed by the model and the counterfactual predictions derived from it.

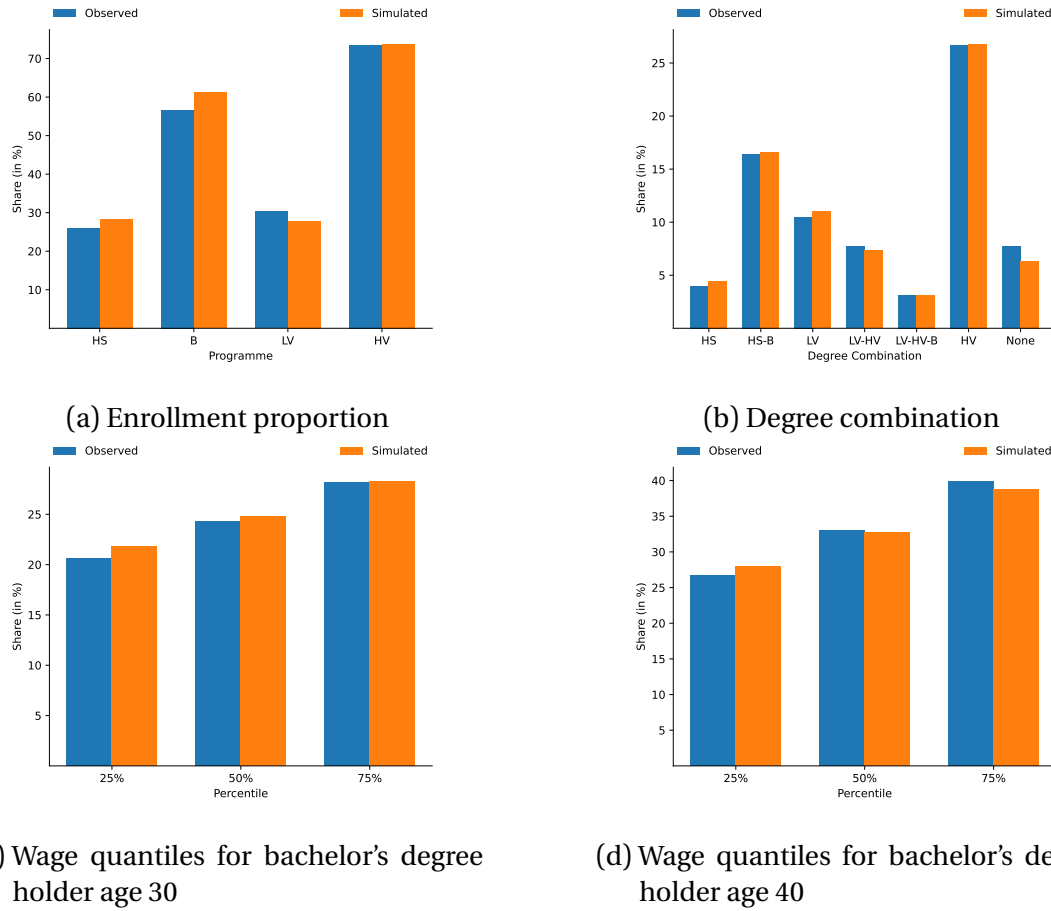


Figure 4: Summary of model fit

Note: This figure summarizes the model fit. The figures compare observed moments based on the dataset described in 2.2 and simulated moments from a model with the estimated parameters. The blue bars show the observed moments, and the orange bars show simulated moments. The x-axis labels for the figures in the first row correspond to education programs specified in Figure 2. AU refers to applied university, AHS refers to applied high school, HV refers to higher vocational program, and LV refers to lower vocational program. Labels in the second figure represent paths through the decision tree specified in Figure 2. AHS-AU, for example, indicates that an individual graduates from applied high school first and from an applied university after that. The figures in the second row depict wage percentiles for individuals who hold a bachelor's degree at ages 30 and 40. In particular, they show the 25th, 50th, and 75th percentiles of wages among all individuals who work full-time and hold an applied university bachelor's degree.

Reform to student income subsidies In 2015, the Dutch government implemented a reform that substantially changed income subsidies for students. The reform removed subsidies for students from high-income families and reduced subsidies for students from low-income families who move out of their parental home. Following the reform, applied university enrollment rates among graduates of vocational programs declined substantially, and common alternative explanations do not account for this decline (Van den Broek et al., 2020). In the [supplemental appendix](#) I show that the decline in enrollment was larger among individuals who are more likely to move out of their parental home, which is consistent with the reform causing the decline in enrollment, as it mainly affected these students.

I compare the model prediction to the effect of the reform by restricting access to the alternative path to university in a way that produces the same drop in enrollment observed in the data. I then compare the change in applied university gradua-

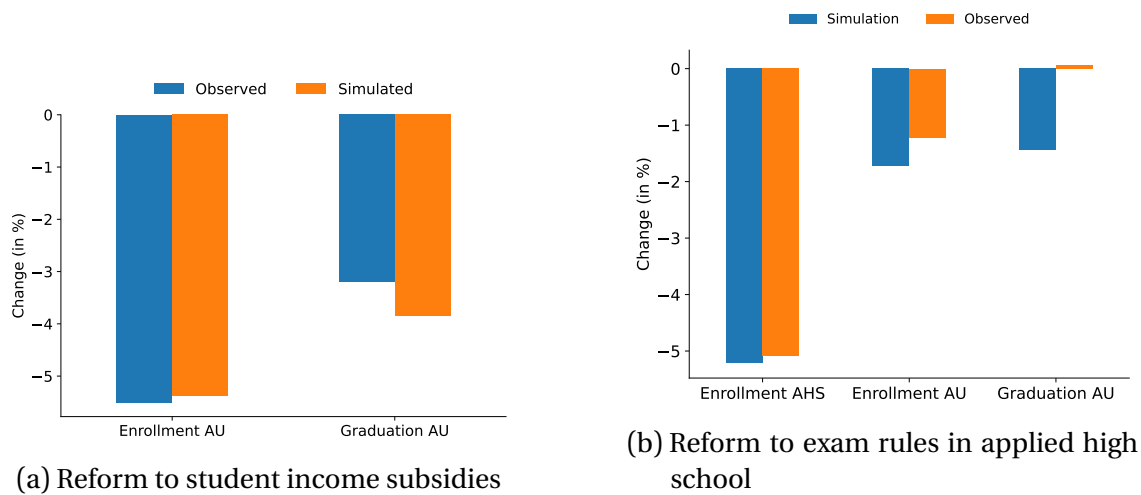


Figure 5: Model Validation

Note: The figure compares the effect of two education reforms to model predictions. The first panel compares the effect of the 2015 reform on graduation rates to model predictions. To obtain the observed data, I compare average enrollment and graduation rates, controlling for composition between the cohorts 2011-2014 and 2015-2017. The simulated changes are derived from a counterfactual simulation that restricts access to the alternative path to university to generate a decline in applied university enrollment similar to the observed decline. The second panel shows a comparison of model predictions to observed changes in applied university and applied high school graduation rates following a reform to exam rules in applied high school. The reform mandated that students need certain minimum grades in a number of core subjects to pass, and has coincided with a decline in transition rates from vocational school to applied high school. The simulated changes are derived from a counterfactual simulation that restricts access to applied high school to generate a decline in applied high school enrollment similar to the observed decline. The orange bar shows the observed change, and the blue bar shows the model prediction. In both panels, the first set of bars is similar by construction as the counterfactuals are designed to match the observed changes in enrollment. AU refers to applied university and AHS refers to applied high school.

tion rates in the data to the change implied by the counterfactual simulation. Figure 5 shows that the model closely predicts the decline in applied university graduation rates following the reform.

Changes to Tracking Rules into Applied High School The second exercise compares model predictions to changes induced by a reform to exam rules in applied high school. Starting with the 2012/2013 cohort, exam rules in applied high school required students to achieve minimum grades in several core subjects to pass. Transition rates from vocational school to applied high school declined substantially after this reform⁵. One possible explanation is that schools became more reluctant to admit graduates from vocational school because they feared lower completion rates. While I cannot rule out other trends that may have contributed to the decline in enrollment, I can examine whether the observed decline aligns with the model predictions.

As before, the model counterfactual is constructed by decreasing the nonpecuniary utility of applied high school by an amount that generates a decline in applied high school enrollment similar to the decline observed in the data. The observed changes are obtained by calculating average enrollment and graduation rates for cohorts 2009-2011 and 2012-2014 as displayed in Figure 5. I then compare the implied

⁵ See Figure S.2 in the [supplemental appendix](#).

changes in applied university enrollment and graduation to the observed changes.

Figure 5 shows that both the counterfactual and the observed data predict a similarly small decline in applied university enrollment in response to the decrease in applied high school enrollment. The model, however, predicts a small decrease in the overall applied university graduation rate, while the observed graduation rate appears unchanged. Graduation in the data is measured by an indicator for whether a person has graduated from applied university ten years after the beginning of the model. This time horizon may not be long enough to capture the decline in graduation, and trends in degree duration may make it difficult to detect the effect in the data. Overall, the model and the observed data agree that the reduction in applied high school transition rates did not lead to a large change in applied university graduation.

5. Results

I now present the empirical findings of the paper. First, I discuss the estimated distribution of wage returns to applied university and dropout risk at applied university. After that, I show several counterfactual simulations derived from the estimated model.

5.1. Mechanisms

The estimated model contains information about the distribution of wage returns to applied university and the distribution of dropout risk at applied university. Both of these factors are crucial in understanding how policy choices shape students' outcomes in the setting of this paper. A detailed list of parameter estimates and standard errors can be found in Section S1 of the [supplemental appendix](#).

Distribution of observed and unobserved characteristics: Individuals are characterized by parental income, academic potential, and a latent type. The joint distribution of parental income and academic potential is observed in the data. The distribution of latent types conditional on parental income and academic potential is estimated along with the other model parameters. Figure 6 shows the joint distribution of fixed characteristics in the model. Consistent with differences in earlier track assignment by socioeconomic status, there are more people from low-income households than people from higher-income households. Latent types are correlated with observed characteristics. Individuals in the lowest academic potential group are more likely to have type θ_5 , whereas individuals in the highest academic potential group are more likely to have type θ_1 . Differences in outcomes across individuals with the

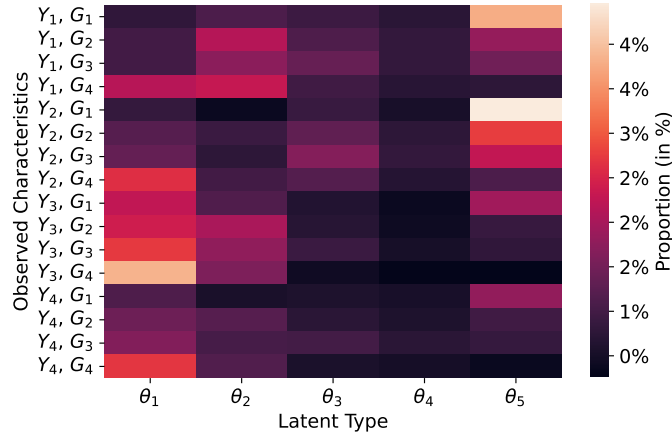


Figure 6: Distribution of fixed characteristics in the model

Note: This figure shows a heatmap visualizing the distribution of fixed characteristics in the model. The vertical axis represents one combination of academic potential and parental income, while the horizontal axis represents one latent type each. Both grades and parental income are exogenously given to the model. The distribution of the latent type given parental income and grades is estimated by the model.

same parental income, ability, and latent type are only due to different realizations of random shocks and are not systematic⁶. If we thus compare wages of individuals with the same parental income, academic potential, and latent type but different education levels, the resulting returns correspond to the average causal wage difference across the education levels for this subgroup. Wage returns conditional on all fixed characteristics are thus average returns to university for all individuals in the respective subgroup. I can thus assign average wage returns to different degrees and average dropout risks in different programs to each individual in the sample.

Wage returns to applied university: The model reveals substantial heterogeneity in wage returns to an applied university degree relative to a higher vocational degree. While average returns at age 40 are positive across the population, they vary sharply by individual characteristics and latent type. Graduates of applied universities receive substantially larger returns to labor market experience than individuals holding higher vocational degrees.

To quantify these differences, I compute the average gap in discounted lifetime income between individuals with an applied university bachelor's degree and those with a higher vocational degree for each combination of observed characteristics and latent type. To construct lifetime earnings, I use the estimated discount rate of around 5% and assume that people receive zero income when they are not working. Furthermore, I assume that wages are constant between the ages of 40 and 70. The results

⁶ The model features school type as an additional fixed characteristic, but since it only affects the utility associated with choices in the first period, it does not directly affect lifetime outcomes.

should be interpreted keeping in mind that these assumptions could lead to a downward bias of lifetime earnings differences if wages continue to grow faster for applied university graduates after age 40.

Figure 7 displays the distribution of lifetime returns. Returns differ most by latent type and much less by parental income or academic potential. The first latent type experiences negative lifetime returns to applied university across most characteristics, whereas the fifth type gains substantially. Figure A.3 shows the same distribution measured at age 40, rather than over the life cycle. At this age, returns are positive for all groups but remain heterogeneous, ranging from roughly 15% for the second latent type to more than 50% for the fifth. For some individuals, the earnings premium later in life is insufficient to offset early career income losses from delayed labor market entry. The model abstracts from job and program-specific factors, such as field of study, or occupation. These omissions may explain why latent type plays such a large role in shaping lifetime income differences across education levels. It is nonetheless remarkable that wage returns do not substantially vary across academic potential.

Figure A.2 shows the returns to an applied university degree relative to all other degrees. These returns are larger than when the alternative is a higher vocational degree, but the distributional patterns are similar. Returns rise when all other degrees are considered as alternative options, since a higher vocational degree is associated with higher wages than a lower vocational degree or dropping out after middle school. Graduates of higher vocational programs also earn slightly more than those with only a high school degree, although this advantage fades over time.

Dropout risk: Heterogeneous dropout risk is the main source of variation in outcomes across individuals in the model. Given that the model predicts substantial returns to applied university for many individuals, a key question is what factors prevent graduation among those without an applied bachelor's degree. Parameter estimates indicate that differences in dropout risk at applied university,⁷ rather than differences in other unexplained preferences, play a particularly important role. To describe how dropout risk is distributed, I calculate the dropout probability at applied university for each combination of observed characteristics and latent type in the model.

Figure 8 shows the resulting distribution. Individuals with lower academic potential face substantial dropout risk and graduate with a probability of about 50%, while those in the highest academic potential group graduate with a probability of around

⁷ The other education programs outlined in Figure 2 also involve dropout risk. I focus on applied university in this section because it is the most relevant program for the long-run outcomes of vocational school graduates. See the supplemental appendix for dropout risk in other programs.

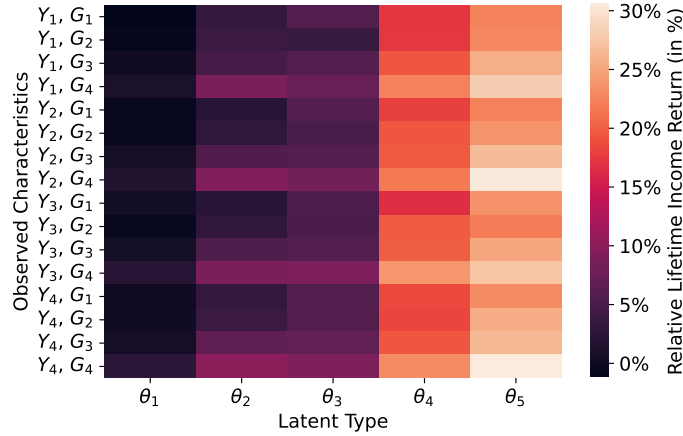


Figure 7: Distribution of lifetime earnings returns to applied university

Note: This heatmap summarizes the distributions of returns to applied universities. The vertical axis represents one combination of academic potential and parental income, while the horizontal axis represents one latent type. The returns are expressed in percentages. The returns are obtained by calculating the difference in discounted lifetime income of individuals with applied bachelor's degrees and of individuals with higher vocational degrees for each combination of observed characteristics and latent type. Observed characteristics are parental income and academic potential. School type is not included since it has no direct effect on wages. Discounted lifetime income differentials within a group of observed characteristics and latent type are average returns to applied university for all individuals in that group since wages don't systematically differ conditional on these variables. The value of the respective group is then assigned to each simulated individual to obtain a distribution.

80%. These differences are consistent with the large gaps in observed dropout rates across quartiles of academic potential. High dropout probabilities for students with low academic potential highlight the importance of providing strong non-academic outside options. The estimated parameters indicate that although students entering university from vocational education are slightly more likely to drop out than those coming from applied high schools, the difference is not particularly large. This finding is noteworthy because it suggests that completing a vocational program that does not explicitly prepare students for applied university and emphasizes practical skills does not substantially reduce the likelihood of succeeding at an applied university.

Unobserved factors also shape dropout risk. Individuals with high expected returns to applied university are also more likely to graduate. Understanding which individuals are affected by a given policy is therefore crucial. If reforms shift mainly those with modest returns and high dropout risk into applied university, the resulting wage effects will be substantially smaller.

5.2. Counterfactuals

In this section, I examine how two policy margins shape students' long-term educational outcomes. The first margin concerns the initial tracking decision, where students enter either an applied high school or a vocational program. The second margin involves the flexibility to pursue higher education after vocational training, specifically the option to enroll in an applied university following a vocational program.

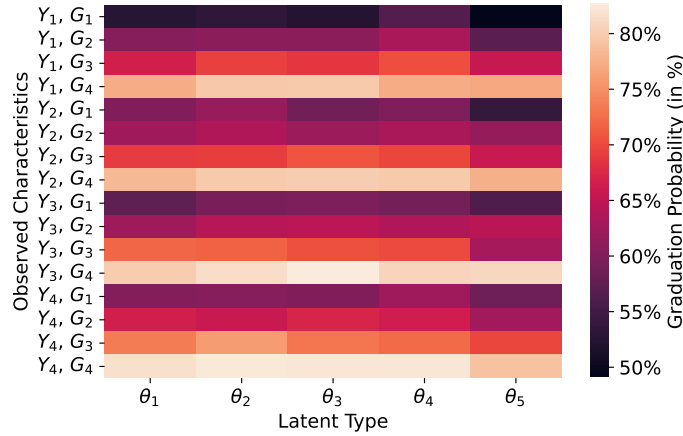


Figure 8: Distribution of graduation probability at applied university

Note: This figure shows a heatmap visualizing the probability of graduation at applied university. The vertical axis represents one combination of academic potential and parental income, while the horizontal axis represents one latent type. This figure is obtained by calculating the number of applied university students who drop out for each combination of observed characteristics and latent type in the model. Observed characteristics are parental income and academic potential. School type is not included since it has no direct effect on wages. Observed dropout rates within a group of observed characteristics and latent type represent academic risk for all individuals in that group since academic risk does not systematically differ conditional on observed characteristics and latent type. The value of the respective group is then assigned to each simulated individual.

These two policies interact. If the main benefit of the alternative path to applied university is that it lowers the cost of entering applied university, then relaxing the initial tracking constraint may be a more effective reform than maintaining a separate alternative route to university. However, if students who later use the alternative path were never close to choosing applied high school initially, preserving access to university through vocational education becomes essential for improving their long-run outcomes. To evaluate these mechanisms, I conduct two sets of counterfactual simulations. First, I simulate policies that expand admission to applied high schools. Second, I simulate policies that restrict access to applied universities for students completing vocational programs. Comparing results across these scenarios isolates the relative importance of early tracking versus flexibility in later educational choices.

Transition costs: Not everyone who graduates from vocational school can easily enroll in applied high school. Applied high schools have different transition rules, so depending on a student's school and grades, it can be harder to enroll. I change two aspects of the model to understand how a more flexible tracking system would affect outcomes. First, I remove differences across school types and simulate a world where everyone faces the rules of the most lenient schools. Second, I reduce transition costs for individuals with lower academic potential because these students face more barriers when moving to high school. Students with lower academic potential have lower grades and usually take an easier final exam. Reducing transition costs for these students represents a world with fewer formal barriers to entering applied high

school.

Both counterfactuals increase applied high school enrollment but have a limited effect on applied university graduation. Figure 9 shows how program enrollment and graduation change across parental income groups. The first panel shows that removing school differences increases applied high school enrollment by about 5 percentage points, while additionally lowering grade barriers increases it by about 15 percentage points across all income groups. The second panel shows smaller effects on applied university enrollment. The first counterfactual raises enrollment by about 1 percentage point, and the second raises it by between 2 and 5 percentage points. The third panel shows that the rise in applied high school graduation is much smaller than the rise in enrollment, especially in the second counterfactual. The fourth panel shows that applied university graduation increases by less than 1 percentage point in the first counterfactual and by about 2 to 4 percentage points in the second.

There are two reasons why the counterfactuals do not significantly increase applied university enrollment, even though they substantially raise applied high school enrollment. First, the counterfactuals shift many students into applied high school who would have pursued the alternative path to university in the baseline scenario. Second, the counterfactuals move students with a high dropout risk at both applied high school and applied university into applied high school. Figure A.5 shows how outcomes vary across quartiles of academic potential, which is particularly important for determining dropout risk in the model. The figure indicates that the first counterfactual mainly shifts students with high academic potential into applied high school. Many of these students would have pursued the alternative university path in the baseline scenario, which explains why the increase in applied university graduation is small. The second counterfactual raises applied high school enrollment across all academic potential groups. However, individuals with lower academic potential experience a smaller increase in applied high school graduation, indicating that a substantial share of them drops out of applied high school.

Figure 10 shows how average wages change across the two counterfactuals. Removing school types increases lifetime wages by about 0.3%, and reducing grade barriers increases them by about 0.5%. The wage effects at age 40 follow a similar pattern. Both policies move some individuals from a higher vocational degree to an applied university degree and shift some from the vocational path to the direct path toward university. Because the direct path is shorter, part of the wage increase comes from greater labor market experience among bachelor's degree holders.

These results show that easing the transition from vocational school to applied high school can improve economic outcomes for some students, but does not lead

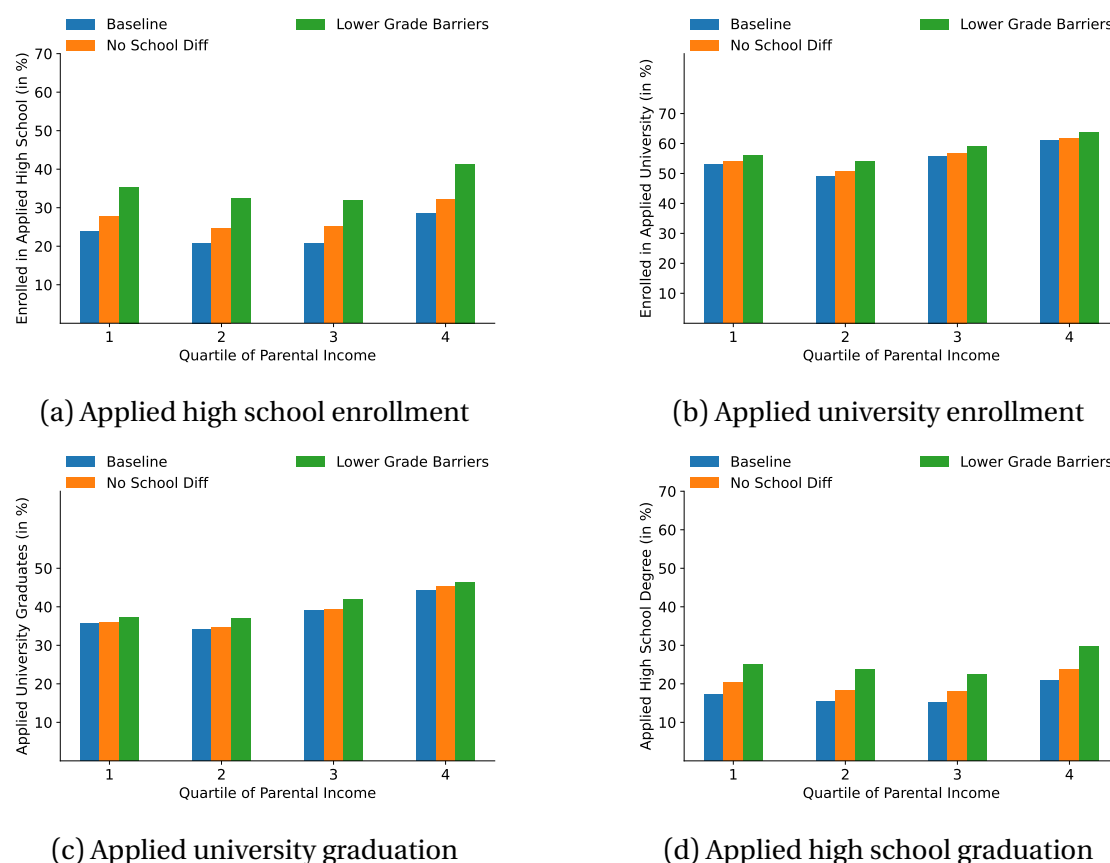


Figure 9: The effect of enforcing higher acceptance rates at high school

Note: This figure shows how enforcing higher acceptance rates at applied high school would affect enrollment and graduation at applied university and applied high school. Blue bars show the baseline model's proportions of applied university and applied high school graduates and enrollees. The orange bars show the proportions in the counterfactual scenario where all schools behave like the most lenient schools. The green bars show the proportions in the counterfactual scenario where people with lower academic potential face lower barriers to enter applied high school, and where all schools behave like the most lenient schools. The proportions are shown for each quartile of parental income. Notably, most individuals graduating from vocational school are from households in the lower-income quartiles.

to large increases in applied university graduation. Removing school types raises life-time wages by increasing graduation and labor market experience and would benefit most affected students. Reducing grade barriers has larger effects on graduation and wages, but these gains must be weighed against higher dropout rates. Most people who drop out of applied high school will enroll in a vocational program, and some will graduate from applied university via the alternative path later in life. This pattern shows how vocational programs and the alternative path to university provide a safety net for students who struggle in applied high school, which dampens the effects of increased dropout rates in applied high school. Expanding access to applied high school helps some students but also draws in others who are not prepared, while some students who would succeed in applied university at a later stage still choose to enroll in vocational programs.

Vocational path to university: In this section, I run two counterfactual simulations to understand the importance of the alternative path to university for wages and

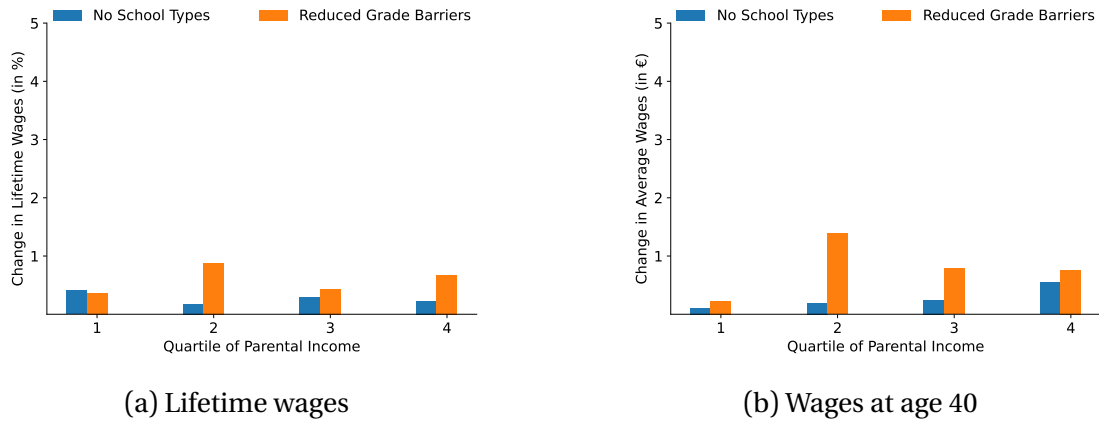


Figure 10: Wage effect of enforcing higher acceptance rates at high school

Note: This figure shows how enforcing higher acceptance rates at applied high school would affect lifetime wages and wages at age 40. The first panel shows changes in average lifetime wages, and the second panel shows changes in average wages at age 40. The blue bar shows wage changes in a counterfactual scenario where each school behaves like the most lenient schools. The orange bar shows wage changes in the counterfactual scenario where people with lower academic potential face lower barriers to enter applied high school, and where all schools behave like the most lenient schools. The changes are shown for each quartile of parental income. Notably, most individuals graduating from vocational school are from households in the lower-income quartiles.

schooling outcomes. The alternative path to university allows people to enter applied university who did not enter applied high school initially, either due to grade barriers or because they were not interested initially. The first motive should disappear if the initial tracking decision were to be made more leniently, while the second effect, being a fundamental consequence of uncertainty in the model, should remain. Thus, I simulate two policies where I restrict access to the alternative path to university while making the initial tracking decision more lenient in the same way as last section. First, I simulate a policy where the alternative path to university takes two years longer than the direct path. This counterfactual is inspired by schooling systems where people need to go back to school to receive some form of high school degree before entering university after finishing a vocational program. The second counterfactual scenario removes the alternative path to university entirely.

Both counterfactuals would lead to a substantial decrease in applied university enrollment and graduation rates. Figure 11 shows how the counterfactual scenarios affect enrollment and graduation at applied high school and applied university across parental income groups. In the first counterfactual, where the alternative path to university is lengthened by two years, applied high school enrollment increases by around 5 percentage points, and in the second counterfactual, where the alternative path to university is removed, applied high school enrollment increases by around 10 percentage points across income groups. Despite the large increase in applied high school enrollment and graduation, both counterfactuals would lead to a substantial decrease in applied university enrollment and graduation. Increasing the length of the alternative path to university by two years would lead to a decrease in applied university graduation rates of around 5 percentage points, and removing the alternative

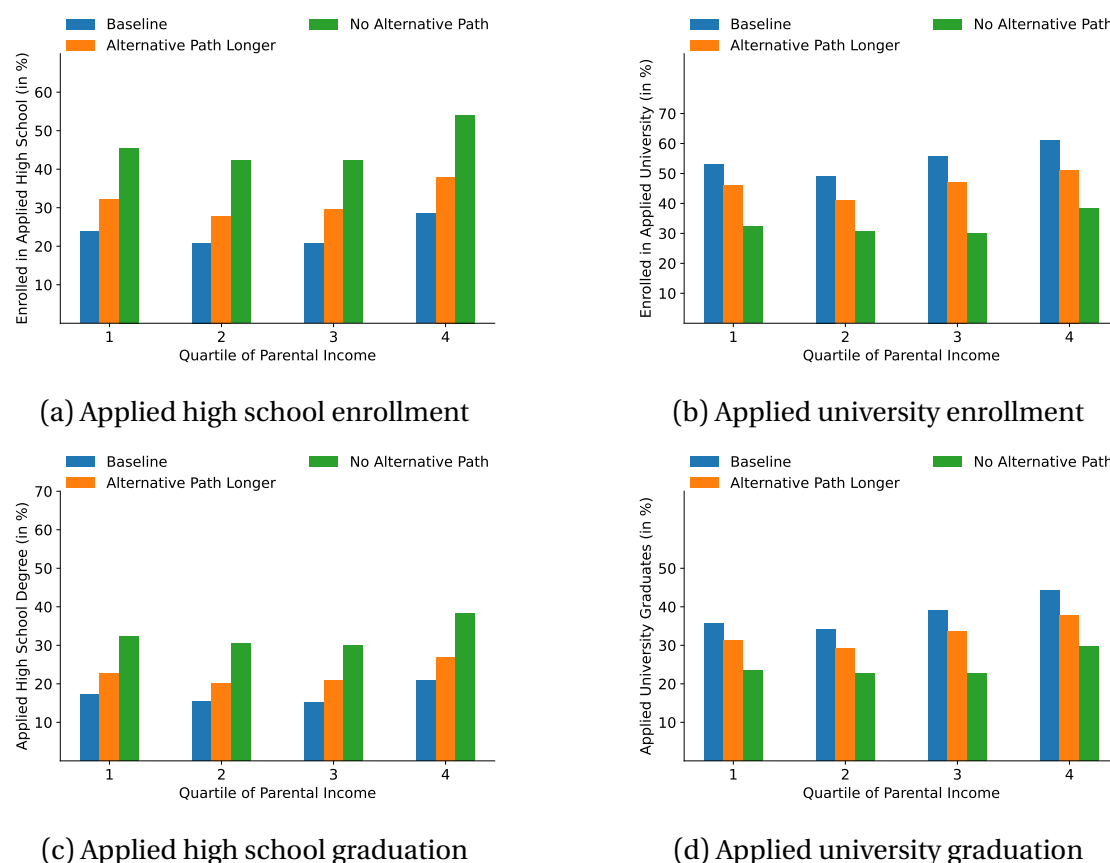


Figure 11: Effect of having no vocational path to applied university

Note: This figure shows how restricting access to the alternative path to university would affect both enrollment and graduation at applied high school and applied university. Blue bars show the baseline model's proportions of applied university and high school-only graduates. The orange bars show the proportions in the counterfactual scenario where the alternative path to university is lengthened by two years. The green bars show the proportions in the counterfactual scenario where people are not allowed to enter applied university after graduating from a higher vocational program. The proportions are shown for each quartile of parental income. In both counterfactual scenarios entry to applied high school is made more lenient according to the counterfactual in the previous section, that removes school types and grade barriers.

path to university entirely would lead to a decrease in applied university graduation rates of about 10 percentage points. In both counterfactuals, the impact of restricting access to the alternative path to university thus dominates the effect of making the initial tracking decision more lenient.

Consistent with the enrollment and graduation effects, both counterfactuals would lead to a substantial decrease in average lifetime wages. Figure 12 shows that both counterfactual simulations would reduce average lifetime income by around 2% across income groups. Removing the vocational path entirely, however, has a much larger impact on wages at age 40, which fall by more than 6%, compared to a 2% decline in the first counterfactual scenario where the program length is increased by two years. The effect on wages at age 40 is larger when removing the alternative path because this leads to a larger reduction in applied university graduation rates and shifts a substantial number of individuals with high expected returns out of applied university. This negative wage effect is partially offset by the fact that many people are shifted

into applied high school who would have taken the alternative path in the baseline scenario. As a result, the overall impact on lifetime wages ends up being similar across the two counterfactuals.

The results of the two counterfactual simulations show that restricting access to the alternative path to university would have substantial negative effects on wages and applied university graduation which cannot be offset by making the initial tracking decision more lenient. Making the tracking system more lenient fails to select a substantial number of people who would benefit from entering applied high school after finishing vocational school. Another pattern that helps to explain the importance of alternative paths to university is that even people who drop out of applied high school enter applied university via the alternative path later in life. These patterns highlight that some people are unprepared to succeed in academic education earlier in their lives, while being able to successfully graduate from applied university later. In such cases, it is impossible to perfectly select people into applied high school at age 15, which provides a rationale for maintaining alternative paths to higher education as fundamental options in the education system.

Taking all of the results together yields three important insights. First, the option to pursue a vocational program improves outcomes for many people who face large dropout risk and modest wage returns to applied university. Second, while it is possible to improve outcomes by making the initial tracking decision more lenient, it is, however, difficult to perfectly differentiate between individuals who would benefit from entering applied high school and those who would not at this stage. Finally, the option to pursue the alternative path to higher education allows people to correct earlier choices after they finish their vocational program. The alternative path to university thus helps to mitigate the negative effects of tracking at age 15 while keeping the benefits of offering different education options.

5.3. Returns to vocational education

Past research has analyzed the returns to vocational education for individuals with different outside options in a fixed institutional setting. Using the model estimated in this paper, I can examine how institutional features influence the earnings effects of vocational education along different margins. In particular, I simulate an expansion of the higher vocational program under both the baseline scenario and a scenario without an alternative path to university. I then compare how outcomes change for individuals who would have chosen a different option in the absence of the expansion in each scenario.

To understand the mechanisms, Figure 13 shows how the expansion affects ed-

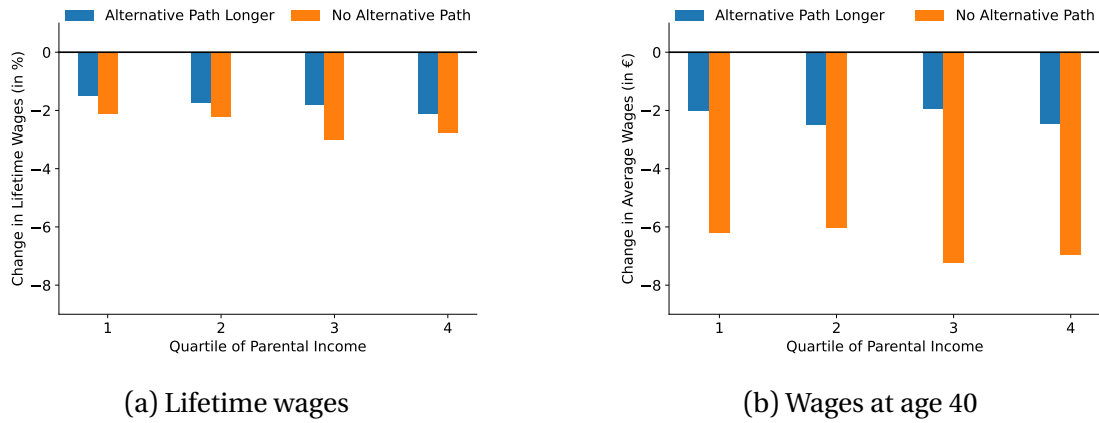


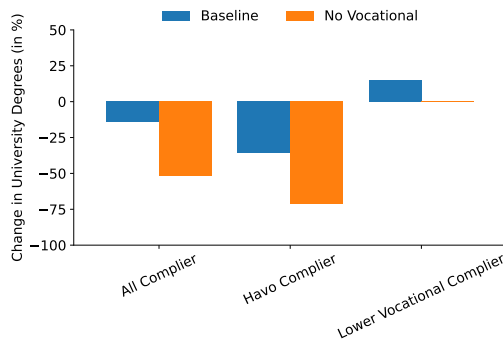
Figure 12: Wage effect of having no vocational path to applied university

Note: This figure shows how restricting access to the alternative path to university would affect lifetime wages and wages at age 40. The first panel shows changes in average lifetime wages, and the second panel shows changes in average wages at age 40. The blue bars show wage changes in the counterfactual scenario where the alternative path to university is lengthened by two years. The orange bars show wage changes in the counterfactual scenario where people are not allowed to enter applied university after graduating from a higher vocational program. In both counterfactual scenarios entry to applied high school is made more lenient according to the counterfactual in the previous section, that removes school types and grade barriers. The changes are shown for each quartile of parental income. Notably, most individuals graduating from vocational school are from households in the lower-income quartiles.

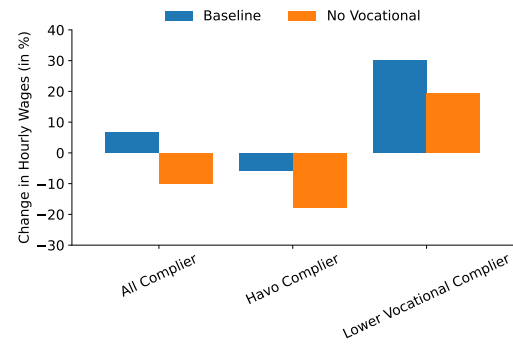
educational outcomes and wages in each scenario. All compliers are individuals whose initial educational choice changes in response to the expansion. HAVO-compliers are individuals who would have chosen the direct academic path to university in the absence of the expansion and instead choose the higher vocational option when the expansion is introduced. This margin corresponds to the diversion margin commonly studied in the literature. MBO3-compliers are individuals who would have chosen the lower vocational program in the absence of the expansion and instead move to the higher vocational program under the expansion. In many previous settings, this margin corresponds to individuals without post-secondary education, but here the relevant outside option is another vocational program due to the more differentiated structure of vocational education in this context.

The figure shows that without the alternative path to university, both complier groups experience substantially lower applied university graduation rates and wages at age 40. In particular, the return to choosing the higher vocational program becomes substantially more negative for HAVO-compliers, indicating that the presence of an alternative path to university can mitigate diversion effects. Similarly, the wage gain for MBO3-compliers is smaller in the scenario without an alternative path to university.

The results show that the returns to vocational programs depend crucially on the institutional setting and the available outside options across different margins.



(a) Applied university graduation



(b) Wages at age 40

Figure 13: Effect of having no vocational path to applied university

Note: This figure shows how returns to a higher vocational program by different alternative choices would change without alternative path to higher education. The figure is obtained by simulating an expansion of the higher vocational program both in the baseline scenario and the scenario without alternative path to university. The expansion is implemented in the model by increasing the utility associated with enrolling in a higher vocational program. HAVO compliers refer to people who would have chosen applied high school in the absence of the expansion. MBO3 compliers refer to people who would have chosen the lower vocational program in the absence of the expansion. The first panel shows the change in university graduation rates among the different complier groups. The second panel shows the change in average hourly wages at age 40 among the different complier groups. The y-axis represents the change in percentage points for both panels. On the x-axis, there are two bars for each of the complier groups. The orange bar shows the return to choosing a higher vocational program in the baseline scenario, while the blue bar shows the same return in the scenario without alternative path to university.

6. Conclusion

In this paper, I have investigated how flexibility in educational choices affects students' outcomes. I estimated a dynamic model of education choices that follows individuals after they graduate from vocational school in the Netherlands. The results show that returns to applied university differ across the population but are substantial for most individuals. I also find that many students face a substantial dropout risk at applied university, creating a wedge between enrollment and graduation rates. Counterfactual simulations show that promoting the direct path to applied university by easing the transition from vocational school to applied high school would benefit some students but would not generate large increases in overall graduation rates or wages. Restricting access to the alternative path to applied university, however, would substantially reduce both graduation rates and wages. The results of this study highlight the importance of both variety in educational options and flexibility in educational pathways for improving students' outcomes in an environment where students are heterogeneous and face uncertainty about their future outcomes.

A. Appendix

A.1. Model parametrization

In this section I show the full model parametrization. Wage equations have been specified in 5 and 6 respectively.

Nonpecuniary returns Formula 17 shows nonpecuniary utility for working without applied university degree. Utility for working with applied university degree looks the same without the degree term.

$$F_v(Y, A_t, E) = \beta_{0,v}^F + \beta_{1,v}^F E_t + \beta_{2,v}^F A_t + \xi_{0,v}^F Y \quad (17)$$

Formula 18 shows nonpecuniary utility for applied university and both forms of vocational training. Utility returns to high school additionally include grades.

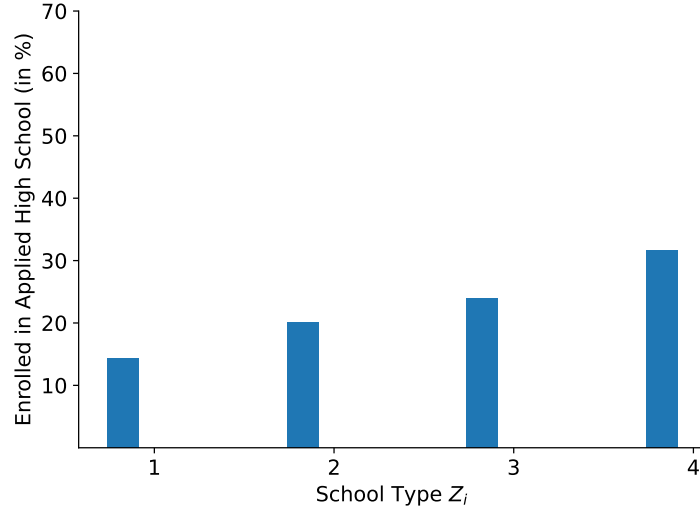
$$F_d(Y, \theta) = \beta_{0,d}^F + \xi_{0,d}^F \theta + \xi_{1,d}^F Y \quad (18)$$

Dropout Risk Formula 2 shows the specification that holds for high school. For university I additionally include an indicator whether an individual has entered university after high school or after vocational training. For the higher vocational program I have left out latent types and for the lower vocational program I have left out both latent types and grades.

Duration Risk Formula 3 shows the specification of duration risk for applied university and higher vocational programs. For the lower vocational program I left out grades. High School and higher vocational training after lower vocational training have fixed lengths.

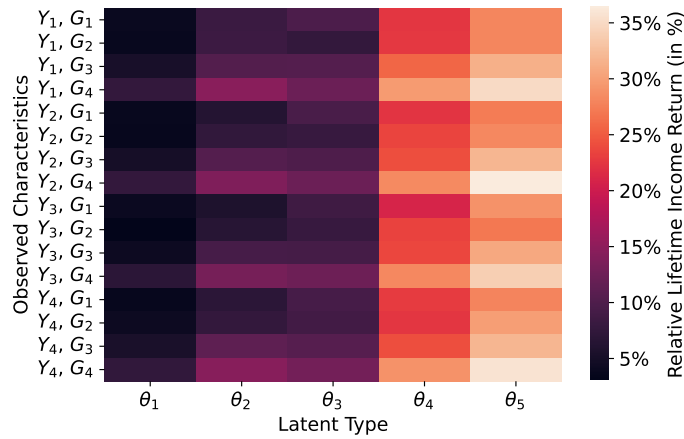
A.2. Additional figures

Figure A.1: Variation in applied high school enrollment rates across school types.



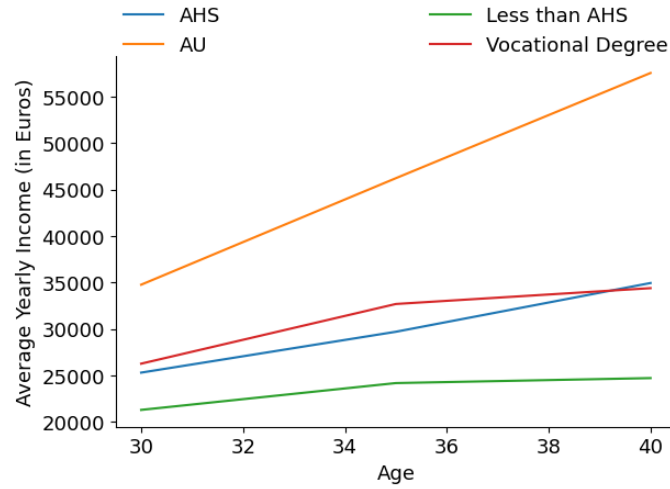
Note: This figure contains the enrollment rates at applied high school across different school types. See Subsection 3.3 for details on how school types are defined.

Figure A.2: Distribution of life-time earnings returns to applied university versus any alternative.



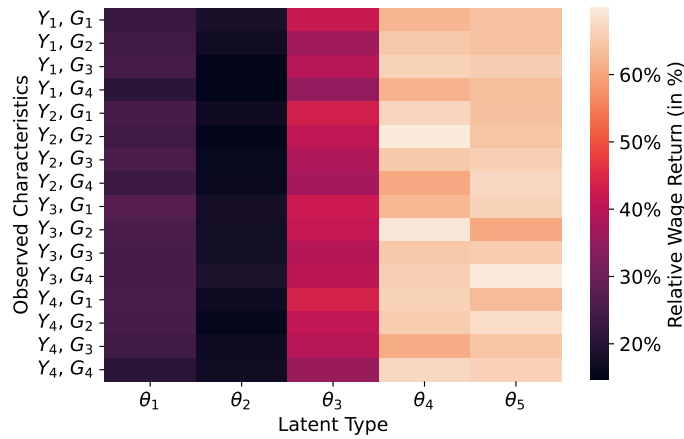
Note: This heatmap summarizes the distributions of returns to applied university versus any alternative. The vertical axis represents one combination of academic potential and parental income, while the horizontal axis represents one latent type. The returns are expressed as percentages. The returns are obtained by calculating the difference in discounted lifetime income of individuals with and without applied university bachelor's degrees for each combination of observed characteristics and latent type. Observed characteristics are parental income and academic potential. School type is not included since it has no direct effect on wages. Discounted lifetime income differentials within a group of observed characteristics and latent type are average returns to applied university for all individuals in that group since wages don't systematically differ conditional on these variables. The value of the respective group is then assigned to each simulated individual to obtain a distribution.

Figure A.4: Wage trajectories by educational attainment

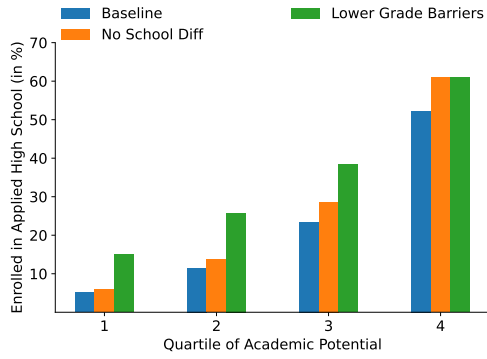


Note: This figure shows the evolution of average hourly wages for individuals with and without applied university degrees. I only include individuals who work full-time. The applied university category only includes individuals with bachelor's degrees. The data is obtained from a cross section of hourly wages in 2019.

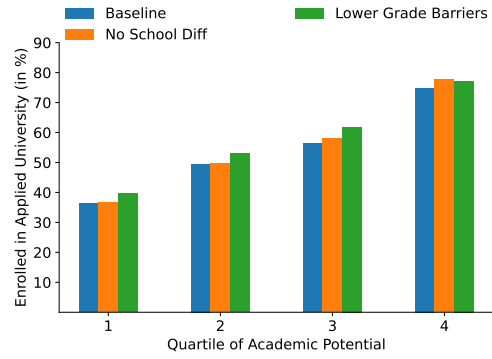
Figure A.3: Distribution of earnings returns to applied university at age 40.



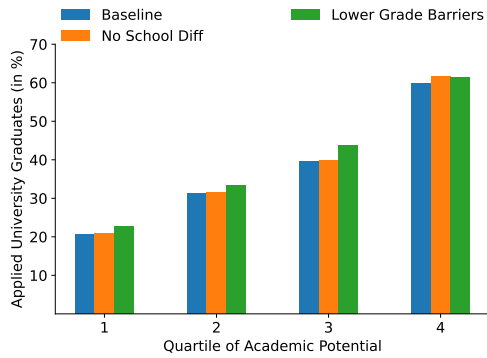
Note: This heatmap summarizes the distributions of returns to applied university at age 40. The vertical axis represents one combination of academic potential and parental income, while the horizontal axis represents one latent type. The returns are expressed in percentage. The returns are obtained by calculating the difference in discounted lifetime income of individuals with applied bachelor's degrees and of individuals with higher vocational degrees for each combination of observed characteristics and latent type. Observed characteristics are parental income and academic potential. School type is not included since it has no direct effect on wages. Discounted lifetime income differentials within a group of observed characteristics and latent type are average returns to applied university for all individuals in that group since wages don't systematically differ conditional on these variables. The value of the respective group is then assigned to each simulated individual to obtain a distribution.



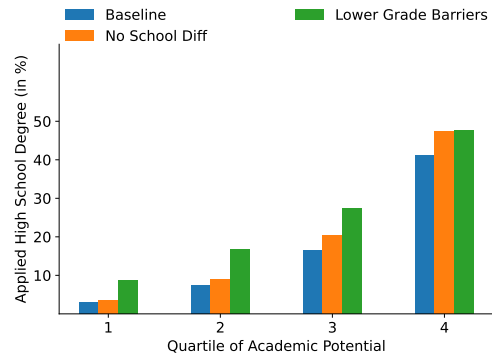
(a) Applied high school enrollment



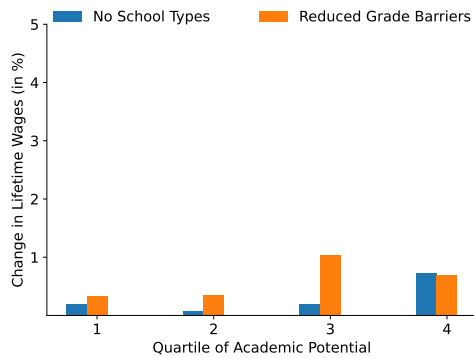
(b) Applied university enrollment



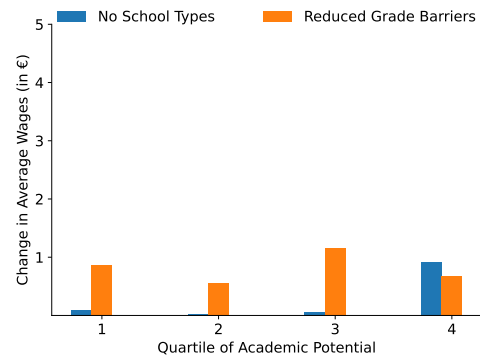
(c) Applied university degree



(d) Applied high school degree



(e) Lifetime wages



(f) Wages at age 40

Figure A.5: The effect of enforcing higher acceptance rates at high school

Note: This figure shows how enforcing higher acceptance rates at applied high school would affect enrollment and graduation at applied university and applied high school. Blue bars show the baseline model's proportions of applied university and applied high school graduates and enrollees. The orange bars show the proportions in the counterfactual scenario where all schools behave like the most lenient schools. The green bars show the proportions in the counterfactual scenario where people with lower academic potential face lower barriers to enter applied high school and where all schools behave like the most lenient schools. The proportions are shown for each quartile of academic potential. The last two panels show the effects on lifetime wages and wages at age 40 respectively.

References

- Adda, J. and Dustmann, C. (2023). Sources of wage growth. *Journal of Political Economy*, 131(2):456–503.
- Altonji, J. G. and Segal, L. M. (1996). Small-sample bias in gmm estimation of covariance structures. *Journal of Business & Economic Statistics*, 14(3):353–366.
- Arcidiacono, P., Aucejo, E., Maurel, A., and Ransom, T. (2016). College attrition and the dynamics of information revelation. Technical report, National Bureau of Economic Research.
- Bertrand, M., Mogstad, M., and Mountjoy, J. (2021). Improving educational pathways to social mobility: evidence from norway’s reform 94. *Journal of Labor Economics*, 39(4):965–1010.
- Birkelund, J. F. and van de Werfhorst, H. G. (2022). Long-term labor market returns to upper secondary school track choice: Leveraging idiosyncratic variation in peers’ choices. *Social Science Research*, 102:102629.
- De Groote, O. (2025). Dynamic effort choice in high school: Costs and benefits of an academic track. *Journal of Labor Economics*, 43(2):467–502.
- Eckardt, D. (2023). Training specificity and occupational mobility: Evidence from german apprenticeships. Technical report, Technical report, mimeo, University of Warwick.
- Fu, C. and Mehta, N. (2018). Ability tracking, school and parental effort, and student achievement: A structural model and estimation. *Journal of Labor Economics*, 36(4):923–979.
- Gabler, J. (2022). A python tool for the estimation of large scale scientific models.
- Hanushek, E. A., Schwerdt, G., Woessmann, L., and Zhang, L. (2017). General education, vocational education, and labor-market outcomes over the lifecycle. *Journal of human resources*, 52(1):48–87.
- Heckman, J. J., Humphries, J. E., LaFontaine, P. A., and Rodriguez, P. L. (2012). Taking the easy way out: How the ged testing program induces students to drop out. *Journal of labor economics*, 30(3):495–520.
- Joensen, J. S. and Mattana, E. (2021). Student aid design, academic achievement, and labor market behavior: grants or loans? *Academic Achievement, and Labor Market Behavior: Grants or Loans*.
- Kasahara, H. and Shimotsu, K. (2009). Nonparametric identification of finite mixture models of dynamic discrete choices. *Econometrica*, 77(1):135–175.
- Keane, M. P. and Wolpin, K. I. (1997). The career decisions of young men. *Journal of political Economy*, 105(3):473–522.

- Matthewes, S. H. and Ventura, G. (2022). On track to success?: Returns to vocational education against different alternatives.
- McNally, S., Ventura, G., and Virtanen, H. (2022). Returns to vocational education and training. In *Handbook of labor, human resources and population economics*, pages 1–30. Springer.
- Mountjoy, J. (2022). Community colleges and upward mobility. *American Economic Review*, 112(8):2580–2630.
- Powell, M. J. et al. (2009). The bobyqa algorithm for bound constrained optimization without derivatives. *Cambridge NA Report NA2009/06*, University of Cambridge, Cambridge, 26.
- Proctor, A. (2022). Did the apple fall far from the tree?
- Silliman, M. and Virtanen, H. (2022). Labor market returns to vocational secondary education. *American Economic Journal: Applied Economics*, 14(1):197–224.
- Stinebrickner, T. and Stinebrickner, R. (2012). Learning about academic ability and the college dropout decision. *Journal of Labor Economics*, 30(4):707–748.
- Tincani, M. M., Kosse, F., and Miglino, E. (2023). *College access when preparedness matters: New evidence from large advantages in college admissions*. Centre for Economic Policy Research.
- Todd, P. E. and Wolpin, K. I. (2023). The best of both worlds: combining randomized controlled trials with structural modeling. *Journal of Economic Literature*, 61(1):41–85.
- Van den Broek, A., Cuppen, J., Ramakers, C., Termorshuizen, T., and Vroegh, T. (2020). Dalende doorstroom mbo-hbo: Waarom stroomt een steeds kleiner aandeel van de mbo-studenten door naar het hbo. *Nijmegen: ResearchNed*.
- Van Esch, W. and J., N. (2010). Van vmbo naar havo: tweestrijd over tweesprong? Technical report.
- Zhu, Z. (2025). Effects of free community colleges: Evidence from a dynamic school choice model. Unpublished manuscript.
- Zumbuehl, M., Chehber, N., Dillingh, R., et al. (2022). Can skills differences explain the gap in track recommendation by socio-economic status? Technical report, CPB Netherlands Bureau for Economic Policy Analysis.